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VISION-BASED RUNNING TECHNIQUE CLASSIFICATION FROM IN-THE-WILD INTERNET VIDEOS USING MARKERLESS POSE ESTIMATION AND DEEP LEARNING

by

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Declaration

I hereby declare that the work presented in this thesis, titled “Vision-Based Running Technique Classification from In-the-Wild Internet Videos using Markerless Pose Estimation and Deep Learning”, is entirely my own original research conducted under the supervision of Nguyen Chi Thien.

All primary and secondary data sources, software tools, literature review materials, and code repositories utilized during this investigation have been appropriately cited and acknowledged in compliance with standard academic protocols. No part of this work has been previously submitted, in whole or in part, for any other degree or qualification at VNUK or any other educational institution.

Date: 10/06/2026 **Candidate’s Signature**

Tran Cong Quoc Huy

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Abstract

This study addresses the problem of vision-based running technique classification utilizing unconstrained, in-the-wild internet videos. Traditional biomechanical analysis relies heavily on marker-based motion capture, which is restricted to controlled laboratory settings. To bridge this research gap, we propose an automated computer vision pipeline that processes raw internet running videos using markerless pose estimation via MediaPipe. The structural framework extracts a custom spatial-temporal signal `contact_diff` through sequential landmark filtering and normalization. An autocorrelation-based automatic gait cycle detection system is established to isolate distinct strides, which are subsequently downsampled into a standardized 8-frame gait phase representation.

The classification task evaluates and compares landmark-based classifiers alongside an image-based MobileNetV2 + LSTM pipeline. To diagnose performance bottlenecks, comprehensive model, signal, and feature ablation studies are performed against a handmade cycle segmentation upper bound and an automatic baseline. While the system demonstrates that markerless pose sequences can effectively encode distinct running-technique characteristics when cycles are isolated cleanly, comprehensive error analysis reveals significant performance degradation in fully automated end-to-end extraction. Specifically, the handmade upper-bound pipeline achieves a test macro-F1 of 0.8991 and an accuracy of 0.9107 on the controlled test split, whereas the fully automatic autocorrelation-based baseline reaches a macro-F1 of 0.5417 and an accuracy of 0.5455. A further 5-fold video-level cross-validation robustness check confirms the same trend under a shared MLP protocol: handmade cycles achieve 0.860 ± 0.016 macro-F1, while autocorr cycles achieve 0.615 ± 0.081, leaving a fair gap of 0.245 macro-F1. A 100-permutation test on the autocorr MLP setting does not reach statistical significance ($p = 0.2475$), suggesting that the autocorr-derived features remain noisy and are not yet robustly separable under a simple classifier. The dataset contains 75 videos and 152 extracted running cycles, but the upper-bound and automatic baseline metrics are reported on their respective controlled test splits rather than on the full dataset. The empirical findings indicate that future deployment readiness depends fundamentally on improving upstream cycle-boundary precision rather than classifier architectural complexity.

Keywords: Markerless Pose Estimation, Action Classification, Gait Cycle Detection, Deep

Learning, Biomechanics.

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Chapter 1

Introduction

1.1 Background

Running technique — the manner in which a person coordinates posture, foot strike, stride, and limb motion through the running cycle — has a direct bearing on athletic performance, training efficiency, and the risk of overuse injury [1]. Coaches, clinicians, and runners therefore have a long-standing interest in assessing whether a given running form is sound or in need of correction. Distinguishing "correct" from "false" technique is, however, a subtle judgement: the diagnostic information is distributed across the whole kinematic chain and is expressed only as the body moves periodically through repeated running cycles, rather than in any single static posture **cutting1978gaitperiodicity**, [2].

Traditional approaches to running analysis are accurate but costly to apply at scale. Optical marker-based motion capture provides laboratory-grade kinematics but requires controlled environments, calibrated multi-camera rigs, and marker placement; wearable inertial sensors enable field measurement but require body-worn instrumentation and calibration; and expert visual inspection depends on scarce specialist time and is inherently subjective [3], [4]. None of these modalities can be applied retrospectively to footage that was not captured with them in mind.

At the same time, *in-the-wild* running videos — clips recorded with ordinary cameras and shared on the internet — are abundant. This footage is an enormous, untapped source of running data, but it is also noisy and uncontrolled: it varies in camera viewpoint, resolution, frame rate, lighting, and framing, and it carries no instrumentation or ground-truth measurement [5]. Recent advances in *markerless pose estimation* make it possible to recover body landmarks directly from such ordinary RGB video, with per-landmark confidence values, and thereby open the possibility of analysing running technique from video that was never intended for biomechanical study [5], [6], [7]. This thesis investigates whether, and under what conditions, that possibility can be realised with deep learning.

1.2 Problem Statement

The problem addressed in this thesis is stated as follows: *given an in-the-wild running video, classify the running technique as correct or false using markerless pose estimation and deep learning.* The input is an ordinary single-runner video with no instrumentation or calibration; the desired output is a binary technique label, ideally accompanied by an indication of confidence. The task must be solved without the controlled conditions assumed by marker-based or sensor-based methods, relying instead on landmarks recovered from the raw footage.

Solving this problem in the wild raises a set of inter-related challenges:

- **Camera viewpoint variation.** Clips are filmed from arbitrary and inconsistent angles, so the same technique appears differently across videos.
- **Uncontrolled resolution and frame rate.** Spatial resolution and temporal sampling (FPS) differ between clips, affecting both pose quality and the temporal granularity of the running cycle.
- **Pose-landmark noise.** Markerless estimates from uncontrolled footage are noisier and less certain than laboratory measurements.
- **Missing or unstable lower-body landmarks.** The ankle, heel, and toe landmarks most diagnostic of running form are also the most frequently occluded, blurred, or mislocalised.
- **Foot-strike and cycle-boundary ambiguity.** The events that define a running cycle are themselves hard to localise reliably from noisy signals.
- **Automatic cycle detection affects classification.** Because the classifier operates on segmented cycles, errors in automatic cycle detection propagate into the final decision, coupling the two stages.

These challenges motivate a design that separates the difficulty of *classification* from the difficulty of automatic *cycle detection*, so that the contribution of each can be measured rather than conflated.

1.3 Research Questions

The thesis is organised around seven research questions:

RQ1. Can pose-based classifiers distinguish correct from false running technique when running cycles are manually or otherwise reliably segmented?

RQ2. How much classification performance is lost when handmade cycles are replaced by automatic autocorrelation-based cycle detection?

RQ3. Which motion signal is most reliable for automatic running-cycle detection?

RQ4. Which model architecture is most suitable for classifying eight-phase pose sequences?

RQ5. How do the feature representation and the number of frames sampled per cycle affect classification performance?

RQ6. How does an image-based representation compare with a landmark-based representation under the same automatic (autocorrelation) pipeline?

RQ7. How do cycle-level aggregation, decision-threshold selection, and pose/cycle quality affect the interpretation of the final results?

RQ1 establishes whether the task is solvable in principle; RQ2 quantifies the cost of automation; RQ3–RQ5 examine the design choices internal to the automatic pipeline; RQ6 addresses the representation question; and RQ7 concerns how results should be measured and interpreted.

1.4 Contributions

In addressing these questions, the thesis makes the following contributions:

1. A complete markerless, pose-based pipeline for classifying running technique from in-the-wild internet videos, spanning pose estimation, landmark filtering and normalisation, motion-signal extraction, running-cycle detection, phase sampling, representation, deep-learning classification, and evaluation.
2. A matched experimental design comprising a *handmade upper bound*, which characterises classification given reliable segmentation, and an *autocorrelation-based automatic baseline*, which uses only information available from the video itself; the pairing isolates the effect of automatic cycle detection.
3. An *eight-phase* running-cycle representation that normalises for cadence so that cycles of differing duration become directly comparable.
4. A set of controlled ablation studies on model architecture, detection signal, feature representation, and frames per cycle.
5. Extended analyses comparing landmark-based and image-based representations on an identical cycle source, comparing cycle-level and video-level evaluation, examining decision-threshold selection, and relating pose and cycle quality to classification accuracy.
6. A disciplined separation between *historical development* results — retained as method-evolution evidence — and *controlled benchmark* results obtained under a single fixed, leakage-free protocol, ensuring that development-stage milestones are not mistaken for benchmarks.

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 reviews the relevant literature on running technique, motion-analysis modalities, markerless pose estimation, running-cycle detection, pose- and image-based classification, and evaluation protocols, and identifies the research gap. Chapter 3 describes the proposed framework in detail, including the dataset and leakage-free splitting protocol, pose estimation, landmark filtering and normalisation, motion-signal extraction, the three cycle-detection methods, eight-phase sampling, the landmark and image representations, the classification models, the cycle- and video-level prediction scheme, and threshold selection. Chapter 4 specifies the experimental setup, including training configurations, evaluation metrics, and the reporting protocol that separates controlled from historical results. Chapter 5 reports the empirical results: dataset statistics, the handmade upper bound, the automatic baseline, the ablation studies, the historical version history, and the extended analyses. Chapter 6 interprets these results, locating where performance is gained and lost across the pipeline. Chapter 7 summarises the findings, restates the answers to the research questions, and outlines directions for future work. Throughout, the scope of the conclusions is kept commensurate with the dataset and protocol used, and no claim of real-world deployment readiness is made.

Chapter 2

Literature Review

This chapter reviews the research relevant to vision-based running-technique classification. It proceeds from the biomechanics of running form, through the sensing modalities used to measure it, to the pose-estimation, cycle-detection, and classification methods on which an automatic pipeline can be built. It then examines how multi-cycle video data should be evaluated, and closes by stating the research gap that this thesis addresses.

2.1 Running Technique Analysis

Running technique, or running form, is commonly characterised by a small set of inter-related kinematic descriptors: the manner of *foot strike* (rearfoot, midfoot, or forefoot contact), trunk and pelvis *posture*, *stride* parameters such as step length and cadence, and the *coordination* of the upper and lower limbs through the gait cycle [1], [2]. These descriptors are not independent; for example, an over-strided rearfoot contact typically coincides with a more extended knee at touchdown and altered trunk lean, so that "correct" and "incorrect" technique are better understood as configurations of the whole kinematic chain than as single isolated events [8].

A recurring theme in the running-biomechanics literature is that technique is expressed *periodically*: the discriminative information lives in how the body moves through repeated running cycles rather than in any single static frame **cutting1978gaitperiodicity**, [2]. This has two consequences for an automatic system. First, the unit of analysis should be a running cycle, not an arbitrary frame window. Second, the descriptors most diagnostic of form — foot contact, knee and ankle behaviour at touchdown, limb swing — are concentrated in the lower body, which motivates particular attention to lower-limb measurement (Section 2.3).

2.2 Marker-Based and Markerless Motion Analysis

Three broad modalities are used to capture running kinematics: optical *marker-based* motion capture, *wearable* inertial sensors, and *markerless* video-based pose estimation. Marker-based systems track reflective markers with calibrated multi-camera rigs and are widely regarded as the laboratory reference for kinematic accuracy, but they require controlled environments, specialist hardware, and marker placement, which makes them unsuitable for analysing the large volumes of uncontrolled footage found online [3]. Wearable inertial measurement units estimate segment motion from accelerometers and gyroscopes; they are portable and usable outdoors, but require body-worn instrumentation, calibration, and drift correction, and are likewise absent from existing internet video [4], [9].

Markerless video-based analysis estimates body configuration directly from ordinary RGB video using learned pose-estimation models. It requires no instrumentation and can in principle be applied to any recording, which is what makes it the only viable modality for *in-the-wild* internet video. The trade-off is reduced and less certain measurement: estimates are noisier, subject to occlusion and viewpoint variation, and accompanied only by model-internal confidence rather than physical calibration [5], [6], [7]. Table 2.1 summarises the comparison that motivates the markerless choice in this thesis.

Table 2.1: Comparison of motion-analysis modalities for running-technique measurement.

Property	Marker-based	Wearable sensors	Markerless video
Instrumentation	Markers + multi-camera rig	Body-worn IMUs	None (RGB video only)
Environment	Controlled lab	Field-capable	Any, incl. in-the-wild
Kinematic accuracy	Reference standard	High, drift-prone	Lower, noisier
Setup / calibration	High	Moderate	None
Applicable to internet video	No	No	Yes
Uncertainty signal	Physical calibration	Sensor model	Per-landmark confidence

The implication is direct: only markerless video can be applied retrospectively to uncontrolled internet footage, so an in-the-wild system must accept the attendant measurement noise and design around it rather than assume laboratory-grade input.

2.3 Human Pose Estimation for Movement Understanding

Modern markerless pipelines rely on human pose estimators that localise a fixed set of anatomical *landmarks* (joints) per frame. For single-person, monocular video, lightweight estimators produce *2D landmarks* — image coordinates for each joint — typically accompanied by a *visibility* or *confidence* value that reflects the model's certainty that the joint is present and correctly located [6], [7], [10]. This per-landmark confidence is central to any downstream use: it allows low-quality detections to be filtered or down-weighted, and it provides a measurable proxy for input quality that can later be related to classification performance.

For running-technique analysis the *lower-body* landmarks — hips, knees, ankles, heels, and toes — are the most informative, because foot strike, knee behaviour at touchdown, and ankle motion are precisely the descriptors that distinguish good from poor form (Section 2.1) [1]. Unfortunately, these same landmarks are also the least *stable*. The feet and ankles are small, fast-moving, frequently occluded by the opposite leg or by motion blur, and often near the image boundary; their estimated positions therefore carry more noise and lower confidence than torso landmarks [3], [5]. This tension — the most diagnostic joints being the least reliably estimated — is a defining challenge of markerless running analysis and recurs throughout this thesis, both in the choice of motion signals (Section 2.4) and in the analysis of error sources.

2.4 Running Cycle and Gait Phase Detection

Because running is periodic, a natural pre-processing step is to segment the continuous pose sequence into individual running cycles, conventionally bounded by successive *foot-strike* events of the same foot (foot-strike-to-foot-strike) [2], [11]. Reliable cycle boundaries are a prerequisite for any phase-aligned representation: if cycles are mis-bounded, samples drawn from "the same phase" across different cycles will in fact correspond to different moments of the gait, undermining comparison.

Two families of method are used to find these boundaries. *Event-based* methods detect foot-strike directly from a chosen motion signal, for instance by locating peaks or valleys in a vertical-displacement or foot-contact proxy signal [11], [12]. *Periodicity-based* methods first estimate the dominant period of the motion — classically via *autocorrelation* of a one-dimensional signal — and then use that period to constrain or validate candidate boundaries, which makes detection more robust to isolated spurious peaks **cutting1978gaitperiodicity**. A practical pipeline often combines the two: an autocorrelation estimate provides the expected cycle length, and peak/valley detection on a foot-contact signal supplies the actual boundaries, with the period acting as a consistency check.

The choice of signal interacts strongly with stability (Section 2.3). Signals derived from torso or hip vertical motion are smooth and easy to detect but only indirectly related to foot strike, while signals derived from the noisy foot and heel landmarks are more directly tied to contact events but more prone to error [12]. Selecting a signal is therefore a trade-off between detection coverage and boundary precision, a trade-off this thesis examines explicitly.

2.5 Pose-Based Action Classification

Once cycles are segmented and sampled, the resulting landmark sequences can be classified by a range of neural architectures, ordered roughly by how much temporal structure they model **jangua2021gait2dposes**, [13]. A *multilayer perceptron* (MLP) treats the flattened sequence as a fixed-length vector and learns a static mapping; it is fast and low-variance but ignores temporal ordering, which can be adequate when inputs are short and well-aligned. A *one-dimensional convolutional neural network* (1D-CNN) applies temporal convolutions across the sequence, capturing local motion patterns with relatively few parameters [14].

Recurrent models capture longer-range temporal dependencies. A *bidirectional long short-term memory* network (BiLSTM) processes the sequence in both directions, allowing each phase to be interpreted in the context of the whole cycle [15], [16]. A *CNN-BiLSTM* hybrid first extracts local features with convolutions and then models their temporal evolution with a BiLSTM, combining local pattern extraction with sequence modelling [17]. More expressive recurrent and hybrid models can represent richer dynamics, but on small or noisy datasets their additional capacity also increases the risk of overfitting — a consideration that is especially relevant when the input cycles are themselves imperfectly segmented.

2.6 Image-Based Video Classification

An alternative to landmark sequences is to classify the *appearance* of the runner directly. Here each sampled phase is represented as an RGB crop centred on the runner, and a convolutional backbone extracts visual features that a temporal model then aggregates over the cycle. A widely used efficient design pairs a lightweight pretrained convolutional backbone such as *MobileNetV2* with a recurrent temporal head such as an LSTM (a MobileNetV2–LSTM model): the backbone encodes each frame into a feature vector, and the LSTM models the sequence of per-frame features to produce a cycle-level prediction [17], [18].

The appeal of image-based representations is that RGB crops retain information absent from sparse landmarks — body silhouette, limb contour, clothing-relative motion, and surrounding visual context — which could in principle aid classification [5]. The cost is a substantially larger and higher-variance input. Image pipelines must learn to ignore appearance variation (background, clothing, lighting) that landmarks discard by construction, which typically demands more training data; they are also sensitive to the stability of the runner-centred crop, which itself depends on the same noisy pose estimates discussed in Section 2.3. Table 2.2 contrasts the two representation families.

Table 2.2: Comparison of landmark-based and image-based cycle representations.

Property	Landmark-based	Image-based (RGB crop)
Input per phase	Joint coordinates (+ confidence)	RGB crop (e.g. 224×224)
Dimensionality	Low	High
Information retained	Joint geometry only	Silhouette, contour, context
Invariance	Appearance-invariant by design	Must be learned
Typical model	MLP / CNN / BiLSTM / CNN-BiLSTM	CNN backbone + temporal head
Data requirement	Lower	Higher
Main vulnerability	Pose-estimation noise	Crop instability, overfitting

2.7 Evaluation Protocols for Multi-Cycle Video Data

A subtlety specific to cycle-based video analysis is the choice of evaluation unit. Because one video can yield several cycles, predictions may be scored at the *cycle level* — treating each detected cycle as an independent sample — or at the *video level* — aggregating a video's cycles into a single prediction by, for example, majority vote, mean probability, or confidence-weighted vote [19]. The two views answer different questions: cycle-level metrics characterise per-cycle discriminability, whereas video-level metrics estimate the performance a user would experience, since the practical unit of interest is the video.

Two methodological hazards accompany this choice. First, if cycles from the same video are allowed to appear in both training and test sets, the evaluation leaks video-specific information and inflates scores; splitting strictly *by video* is therefore essential. Second, comparisons across pipelines are only fair when the evaluation unit and protocol are held fixed; a model evaluated on whole-video sequences is not directly comparable to one evaluated per cycle. Sound practice is to report the split protocol explicitly, to fix the evaluation unit within any comparison, and to state whether a tuned decision threshold was used and how it was selected.

2.8 Research Gap

The reviewed literature establishes that running technique is a periodic, lower-body-dominated phenomenon (Section 2.1); that only markerless video can be applied to in-the-wild footage, at the price of noisy estimates (Section 2.2); that the most diagnostic landmarks are also the least stable (Section 2.3); and that cycle detection, representation, and classification each offer well-studied options (Sections 2.4–2.6). What remains under-addressed is their combination on *uncontrolled internet video* under a rigorous, leakage-free evaluation.

Specifically, three gaps motivate this thesis. First, prior work seldom *separates* the contribution of automatic cycle detection from that of the classifier; without a controlled upper bound, it is unclear whether failures stem from segmentation or from learning. Second, landmark-based and image-based representations are rarely compared on an *identical* automatic cycle source under a fixed split, so apparent representation advantages may merely reflect protocol differences — and historical, differently-evaluated milestones risk being mistaken for controlled benchmarks. Third, the interaction between pose/cycle quality and classification accuracy on in-the-wild data is rarely quantified, leaving the dominant error source unidentified. This thesis addresses these gaps with a matched handmade-upper-bound versus automatic-baseline design, a controlled landmark-versus-image comparison on a shared cycle source, and an explicit analysis of cycle-level versus video-level evaluation, threshold selection, and pose/cycle quality (Chapters 3–6).

Chapter 3

Methodology

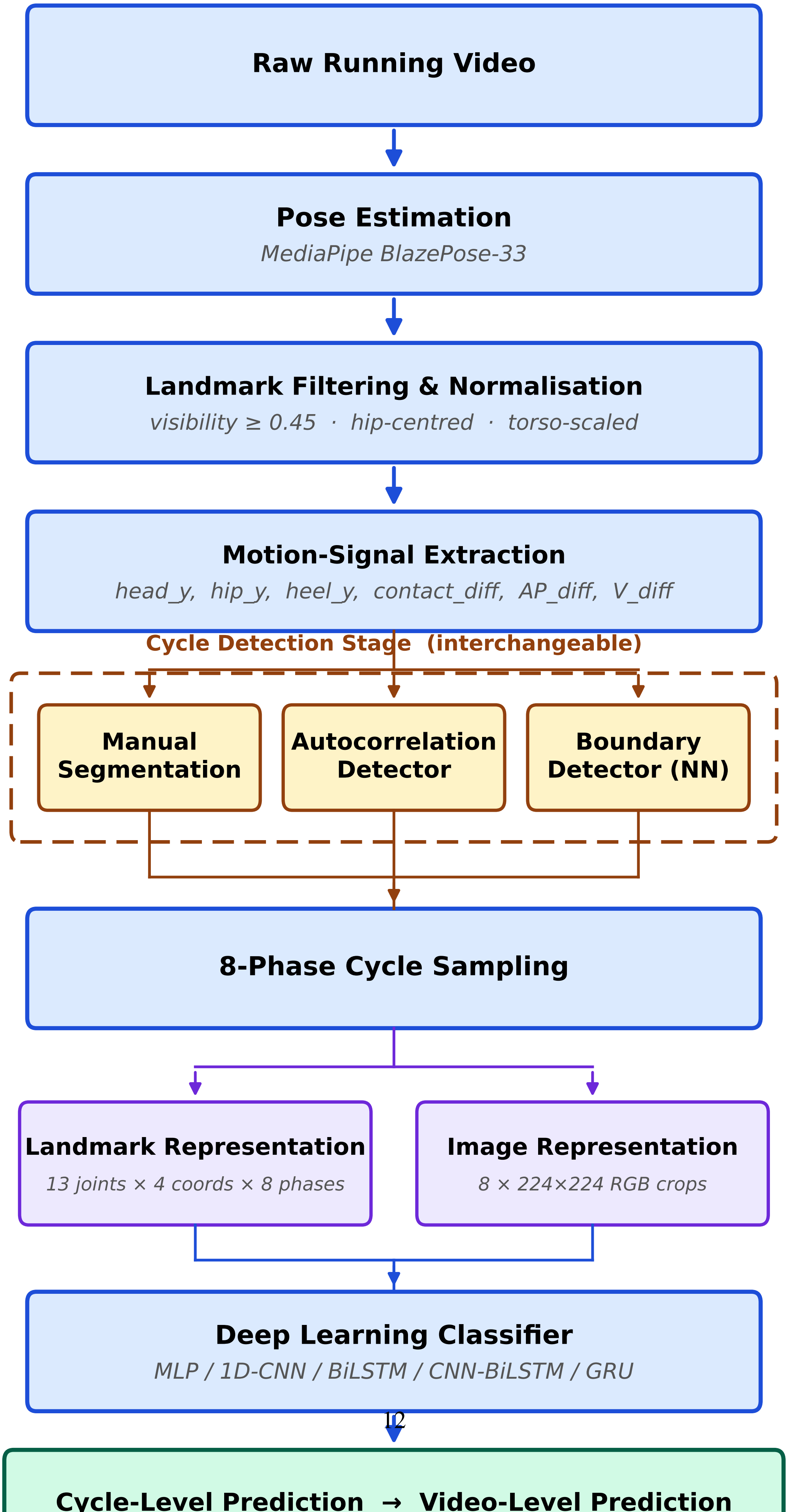
This chapter describes the proposed framework for classifying running technique from in-the-wild video. It follows the pipeline end to end: pose estimation, landmark filtering and normalisation, motion-signal extraction, cycle detection, eight-phase sampling, the two cycle representations, the classification models, the cycle- and video-level prediction scheme, and threshold selection. No results are reported; the experimental outcomes appear in Chapter 5.

3.1 Overview of the Proposed Framework

The framework transforms a raw video into a correct/false technique prediction through the stages shown in Figure 3.1: markerless pose estimation, landmark filtering, motion-signal extraction, running-cycle detection, eight-phase sampling, construction of a pose-based or image-based representation, deep-learning classification, and evaluation. A deliberate design principle is to keep the cycle-detection stage interchangeable so that a manual (handmade) segmentation, a rule-based autocorrelation detector, and a learned boundary detector can be compared under an otherwise identical pipeline, isolating the effect of cycle source on performance.

3.2 Dataset Construction and Splitting Protocol

The dataset consists of single-runner clips labelled as correct or false technique. Because a single video yields multiple running cycles, the *splitting protocol* is critical: the data are split *by video identifier*, so that all cycles extracted from a given video are confined to a single partition (train, validation, or test). This prevents *leakage* of video-specific information — if cycles from one video appeared in both training and test sets, the model could recognise the video rather than the technique, inflating the apparent accuracy. The split is fixed once and reused unchanged across every experiment so that all pipelines are evaluated on the same videos. The resulting per-split counts are reported in Chapter 5.


Raw Running Video
Pose Estimation
MediaPipe BlazePose-33
Landmark Filtering & Normalisation
visibility ≥ 0.45 · hip-centred · torso-scaled
Motion-Signal Extraction
head_y, hip_y, heel_y, contact_diff, AP_diff, V_diff
Cycle Detection Stage (interchangeable)
Manual Segmentation
Autocorrelation Detector
Boundary Detector (NN)
8-Phase Cycle Sampling
Landmark Representation
13 joints × 4 coords × 8 phases
Image Representation
8 × 224×224 RGB crops
Deep Learning Classifier
MLP / 1D-CNN / BiLSTM / CNN-BiLSTM / GRU
Cycle-Level Prediction → Video-Level Prediction

3.3 Markerless Pose Estimation

Each frame is processed by a markerless pose estimator that returns a fixed set of body landmarks with image coordinates and a per-landmark *visibility* (confidence) value [6], [7]. The visibility value is retained throughout the pipeline: it is used both to filter unreliable detections (Section 3.4) and, in later analysis, as a measurable proxy for input quality. A reduced set of *selected landmarks* relevant to running technique — nose, shoulders, hips, knees, ankles, heels, and foot indices — is used as the basis of the pose representation, since the discriminative information for running form is concentrated in the trunk orientation and the lower limbs (Chapter 2). The lower-limb landmarks (ankles, heels, toes) are the most diagnostic but also the most prone to noise and occlusion, which motivates the filtering and normalisation described next. Figure 3.2 shows an example of the estimator output on a running frame.

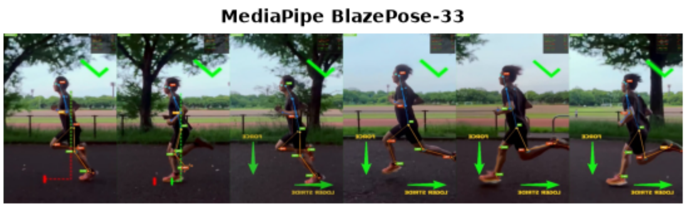


Figure 3.2: MediaPipe BlazePose-33 applied to a sample running frame. The model produces 33 body landmarks with per-landmark visibility scores. A subset of 13 landmarks relevant to running form — nose, hips, knees, ankles, heels, and foot indices — is selected for all downstream processing.

3.4 Landmark Filtering and Normalization

Raw landmark trajectories are filtered before use. Detections whose visibility falls below a threshold are treated as unreliable and discarded or interpolated, and the resulting per-joint trajectories are temporally smoothed to suppress high-frequency jitter introduced by frame-to-frame estimation noise. Visibility-based screening is applied with particular care to the nose and heel landmarks, which serve as quality indicators for the head and foot regions respectively.

Landmarks are then *normalised* so that the representation is invariant to the runner's position and apparent scale in the frame. Coordinates are expressed relative to a body-centred reference and scaled by a body-size measure (for example a torso-span normalisation), removing dependence on where the runner appears in the image and how far they are from the camera. This normalisation is essential for in-the-wild video, where framing, zoom, and runner position vary arbitrarily between clips. Figure 3.3 illustrates the effect on the vertical trajectories of the nose, hip, and heel landmarks.

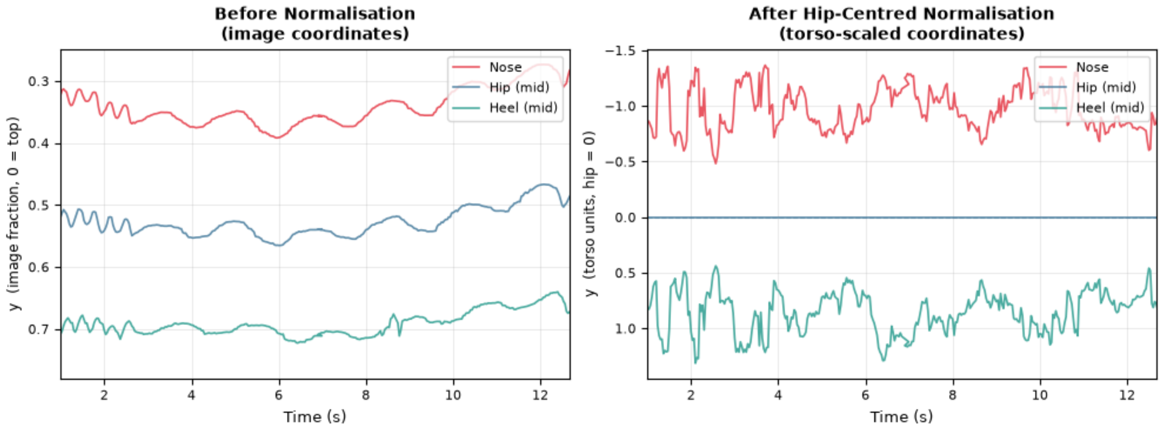


Figure 3.3: Landmark vertical coordinates before (left) and after (right) hip-centred normalisation for an example video. Before normalisation, the trajectories drift with the runner's absolute position in the frame. After normalisation, all trajectories oscillate stably about zero (the hip reference) with consistent amplitude regardless of camera distance or runner position.

3.5 Motion Signal Extraction

From the filtered, normalised landmarks, a set of one-dimensional *motion signals* is derived for cycle detection. Each signal is a scalar time series whose periodic structure reflects the running cycle. Table 3.1 lists the candidate signals.

Table 3.1: Candidate one-dimensional motion signals used for running-cycle detection.

Signal	Definition (per frame)
head-Y	Vertical position of the head/nose landmark
hip-Y	Vertical position of the hip (pelvis) landmark
heel-Y	Vertical position of the heel landmark
toe-Y	Vertical position of the toe (foot index) landmark
`contact_diff`	Foot-contact proxy from the heel/toe landmarks (left–right contact difference)
`AP_diff`	Anterior–posterior (horizontal) displacement difference between feet
`V_diff`	Vertical displacement difference between feet

The vertical-oscillation signals (head-Y, hip-Y) are smooth and easy to detect but only indirectly related to foot strike, whereas the foot-derived signals (`contact_diff`, `AP_diff`, `V_diff`, heel-Y, toe-Y) are more directly tied to ground contact but noisier (Chapter 2). The pipeline supports any single signal as the detection driver; the comparison of signals is treated as an ablation (Chapter 5). Each signal is smoothed and detrended before detection to stabilise period estimation. Figure 3.4 shows the four main signals over an example running clip.

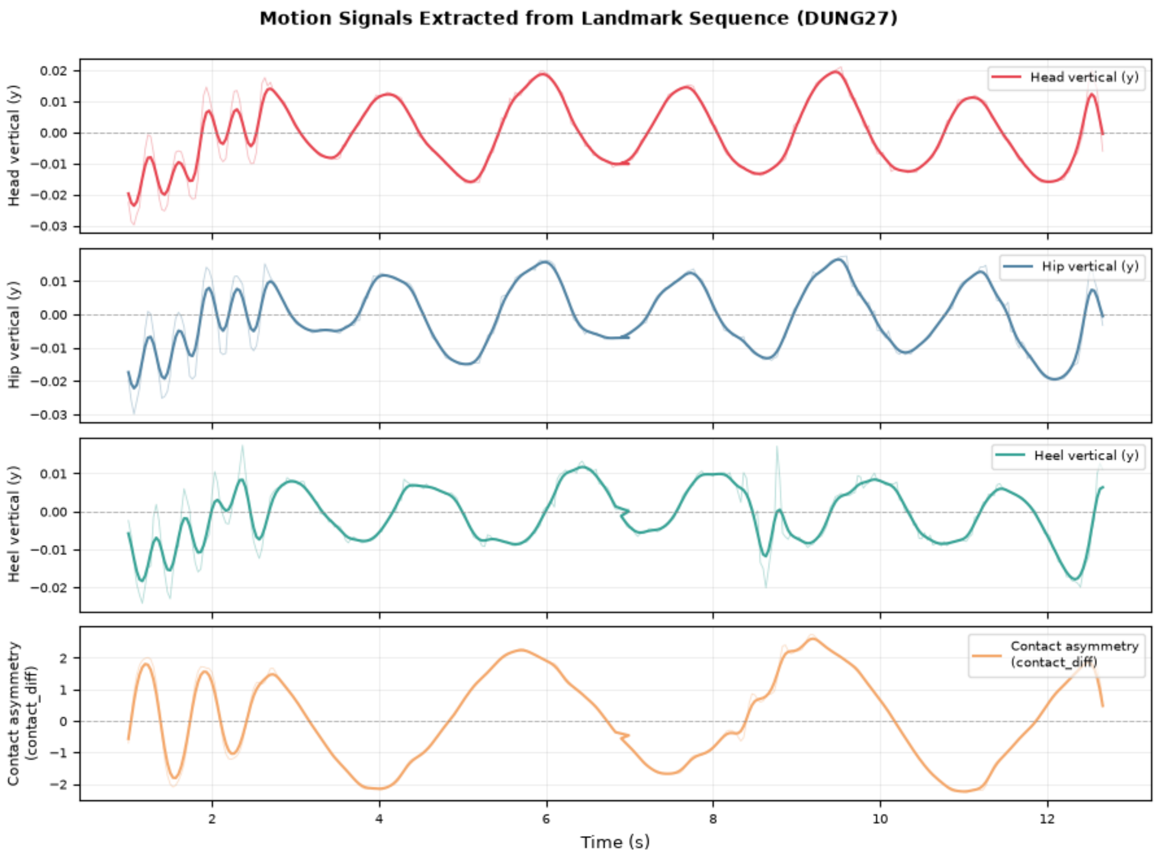


Figure 3.4: Motion signals extracted from a single running video (DUNG27). Faint lines show the raw detrended signal; solid lines show the Gaussian-smoothed version ($\sigma = 2$). The periodic oscillation visible in all four signals reflects the running cycle; `contact_diff` captures left–right foot-contact asymmetry and is used as the primary cycle-detection driver.

3.6 Running Cycle Detection

A running cycle is bounded by successive foot-strike events of the same foot (foot-strike-to-foot-strike, FS-to-FS). The framework provides three interchangeable detectors — a manual handmade segmentation, an automatic autocorrelation-based detector, and a learned boundary detector — followed by a common quality-filtering step.

3.6.1 Handmade Cycle Segmentation

In the handmade pipeline, cycle boundaries are determined manually so that each extracted cycle is correctly segmented. This pipeline is used as an *upper bound*: it characterises how well the representation and classifier can separate correct from false technique *when segmentation is reliable*, and thereby separates the difficulty of classification from the difficulty of automatic detection. It is not a deployable baseline, since a fielded system cannot rely on manual segmentation.

3.6.2 Autocorrelation-Based Automatic Cycle Detection

The automatic baseline detects cycles without manual input. A chosen motion signal is smoothed and detrended; its dominant period is estimated by *autocorrelation*; candidate foot-strike boundaries are located by peak/valley detection on the signal; and the estimated period is used to validate boundary spacing, including a correction for spurious double-stride detections (a two-stride fix). Candidate cycles are then passed to quality filtering. Algorithm 1 summarises the procedure.

Algorithm 1 Autocorrelation-based running cycle detection

Require: Filtered landmark sequence, motion signal s, duration bounds $[\tau_{\min}, \tau_{\max}]$
Ensure: Set of accepted FS-to-FS running cycles C

$x \leftarrow \textsc{ExtractSignal}(s)$
$x \leftarrow \textsc{Smooth}(x)$
$\hat{p} \leftarrow \textsc{EstimatePeriod}(\textsc{Autocorrelation}(x))$
$P \leftarrow \textsc{FindCandidateBoundaries}(x)$
$P \leftarrow \textsc{TwoStrideFix}(P, \hat{p})$
$C \leftarrow \emptyset$
for each consecutive boundary pair (f_i, f_{i+1}) in P **do**
    $\tau \leftarrow \textsc{Duration}(f_i, f_{i+1})$
    if $\tau_{\min} \leq \tau \leq \tau_{\max}$ and visibility is adequate **then**
        $C \leftarrow C \cup \{(f_i, f_{i+1})\}$
    end if
end for
return $\textsc{QualityFilter}(C)$

This detector is the main *automatic baseline* of the thesis: it uses only information available from the video itself, with no fixed-margin assumption on boundary placement, and produces cycles directly comparable to the handmade ones because the downstream stages are identical. Figure 3.5 shows an example detection result.

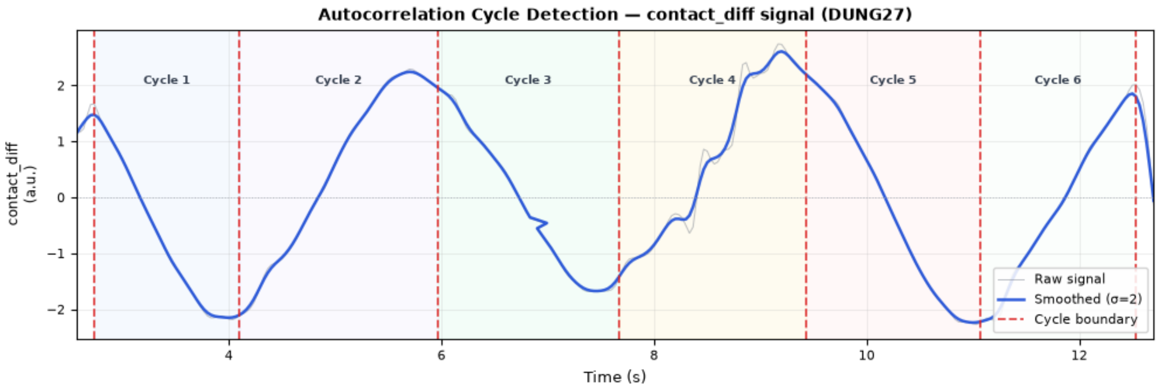


Figure 3.5: Running cycle boundaries detected by the autocorrelation-based detector overlaid on the `contact_diff` signal for an example video (DUNG27). Alternating shaded bands correspond to individual FS-to-FS cycles; red dashed lines mark the detected boundaries. The periodic structure of the signal makes the boundaries reliably locatable by the autocorrelation procedure.

3.6.3 Boundary Detector-Based Cycle Detection

As a learned alternative to rule-based detection, a *boundary detector* (BD) replaces autocorrelation with a trained sequence model [12], [20]. The BD is a BiLSTM that consumes a sliding window of frames and predicts the probability that the central frame is a foot-strike boundary; ground-truth boundaries are taken from the endpoints of handmade single-cycle clips. At inference, the per-frame boundary probabilities are smoothed and thresholded, peaks are taken as boundaries, and consecutive boundaries define FS-to-FS cycles, which are then quality-filtered exactly as for the autocorrelation detector. The BD is treated as a supplementary pipeline.

3.6.4 Cycle Quality Filtering

Regardless of detector, candidate cycles pass through a common *quality-filtering* step before being accepted. Cycles whose duration falls outside the plausible running-cycle range, whose constituent landmarks have insufficient visibility, or which cannot be sampled into the required number of phases are rejected. This step enforces a consistent definition of a usable cycle across all pipelines, so that the only intended difference between them is how boundaries are obtained. The rejected-cycle statistics are later used to analyse where the automatic pipeline loses data (Chapter 5).

3.7 Eight-Phase Cycle Sampling

Each accepted cycle is resampled into a fixed number of *gait phases* so that cycles of different absolute durations become directly comparable. Eight phases are used, spanning one full FS-to-FS cycle:

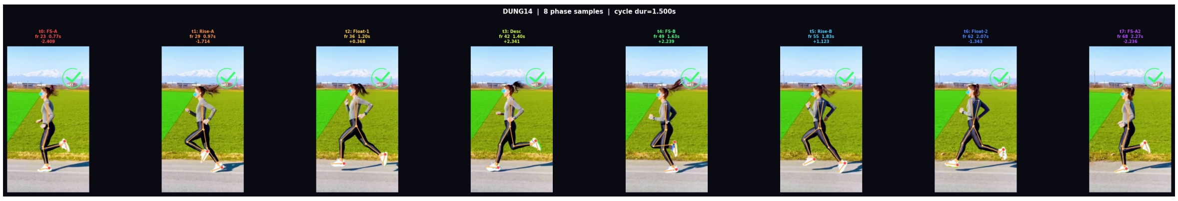


Figure 3.6: Example of eight-phase sampling from an automatically detected running cycle.

Sampling at fixed phases (rather than fixed frame offsets) normalises for cadence, so that "the same phase" denotes the same point in the running cycle across runners and speeds [2]. The eight-phase sampling defines the temporal length of every representation that follows; its sufficiency relative to finer samplings is examined as a frames-per-cycle ablation in Chapter 5.

3.8 Landmark-Based Pose Representation

In the landmark representation, each of the eight phases is encoded by the normalised coordinates of the selected landmarks, optionally augmented with *velocity* (the inter-phase change in coordinates) and other derived quantities. The result is a compact, fixed-size sequence per cycle that is invariant to appearance by construction. This representation is the basis of the main automatic baseline and of the model, feature, and signal ablations.

3.9 Image-Based Cycle Representation

In the image representation, each phase is encoded as an RGB *crop* centred on the runner. A stable bounding box is computed per cycle from the pose estimates (for example a median box over the cycle) and used to crop and resize each sampled frame to a fixed resolution, yielding an eight-frame image sequence per cycle. This representation retains appearance information absent from landmarks — silhouette, limb contour, clothing-relative motion — at the cost of higher dimensionality and sensitivity to crop stability. To ensure a *fair* comparison, the image pipeline uses exactly the same automatic cycle source as the landmark baseline, so that the two differ only in representation (Chapter 5).

3.10 Classification Models

Several classifiers are used across the representations and pipelines, chosen to span a range of temporal modelling capacity. Figure 3.7 shows the layer-by-layer architectures of the four landmark-based models.

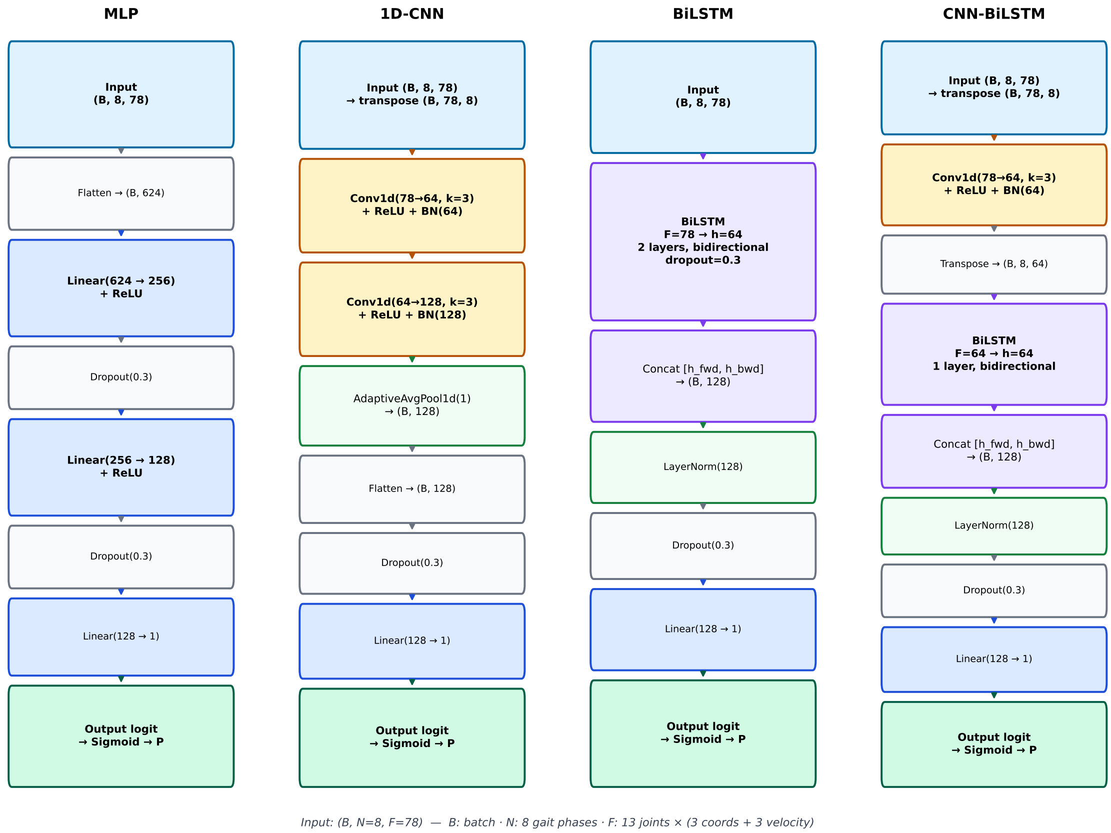


Figure 3.7: Layer-by-layer architectures of the four landmark-based classifiers. All models receive an input tensor of shape $(B, 8, 78)$ — batch size, eight gait phases, and F=78 features (13 joints × 3 coordinates + 3 velocity components) — and produce a single binary logit. Colour indicates layer type: blue — linear; amber — convolutional; violet — recurrent (BiLSTM); green — normalisation or pooling; grey — utility (flatten, dropout).

3.10.1 MLP Baseline

The multilayer perceptron flattens the eight-phase landmark sequence into a single vector and learns a static mapping to the correct/false decision. It does not model temporal order, but is fast and low-variance, which can be advantageous for short, well-aligned inputs.

3.10.2 One-Dimensional CNN

The 1D-CNN applies temporal convolutions across the eight-phase sequence, capturing local kinematic patterns with relatively few parameters before a classification head [14].

3.10.3 BiLSTM

The bidirectional LSTM processes the phase sequence in both directions, so that each phase is represented in the context of the entire cycle, capturing longer-range temporal dependencies than the convolutional model [15], [16].

3.10.4 CNN-BiLSTM

The CNN-BiLSTM hybrid first extracts local features with one-dimensional convolutions and then models their temporal evolution with a BiLSTM, combining local pattern extraction with sequence modelling [17].

3.10.5 MobileNetV2-LSTM

For the image representation, a MobileNetV2–LSTM model is used: a MobileNetV2 convolutional backbone encodes each RGB crop into a feature vector, and an LSTM aggregates the eight per-phase feature vectors into a cycle-level prediction. The backbone may be frozen or partially fine-tuned [17], [18]. This model operates on image crops rather than landmark coordinates and is therefore used only in the image-based pipeline.

3.10.6 GRU for Raw Sequence Modeling

To probe whether explicit cycle cutting is necessary, a recurrent model (a gated recurrent unit, GRU, alongside a BiLSTM) is applied to the *raw* pose sequence of an entire video, without segmentation into cycles. Here one sample corresponds to one video, so this is an exploratory, supplementary configuration rather than part of the cycle-based comparison.

3.11 Cycle-Level and Video-Level Prediction

Because each video yields several cycles, predictions are produced at two levels. *Cycle-level* prediction treats each detected cycle as an independent sample. *Video-level* prediction aggregates a video's cycle predictions into a single decision; three aggregation rules are considered: *majority vote*, *mean probability*, and *confidence-weighted vote*. Since the underlying data unit is the video, video-level prediction is the more faithful measure of practical performance; both levels are reported so that the effect of aggregation can be examined (Chapter 5).

3.12 Threshold Selection

Each probabilistic classifier requires a *decision threshold* to convert predicted probabilities into a correct/false label. Two settings are considered: the default threshold of 0.5, and a *tuned* threshold selected on the *validation* set by sweeping candidate thresholds and choosing the one that maximises validation macro-F1, which is then applied unchanged to the test set. Selecting the threshold on validation rather than on test avoids optimistic bias; reporting both the default and the tuned result makes any threshold overfitting visible. Every reported metric states which threshold setting it uses.

3.13 Implementation Outputs

The pipeline is implemented as a sequence of self-contained, configurable notebooks, each writing standardised outputs — per-cycle and per-video prediction files, result summaries, cycle-detection statistics, and diagnostic figures — to a fixed directory structure. These artefacts provide the inputs to the controlled comparisons, ablations, and quality analyses presented in Chapter 5, and are designed so that missing files degrade gracefully rather than invalidating downstream aggregation.

Chapter 4

Experimental Setup

This chapter specifies how the framework of Chapter 3 is evaluated: the fixed dataset split and evaluation protocol, the experimental pipelines and their roles, the model training configuration, the evaluation metrics, and the design of the ablation and extended analyses. No results are reported here; all empirical outcomes appear in Chapter 5.

4.1 Dataset Split and Evaluation Protocol

All experiments use a single *fixed split by video identifier*. Each video is assigned to exactly one partition — train, validation, or test — and all running cycles extracted from that video inherit its partition. This guarantees that no cycle from a test video is ever seen during training, preventing the model from recognising a specific video rather than the technique and thereby avoiding the optimistic bias that video-level leakage would introduce . The split is generated once and reused unchanged by every pipeline and every ablation, so that all configurations are compared on identical videos.

The validation partition is used for model selection and for decision-threshold tuning (Section 4.4 and Section 4.5.7); the test partition is held out and used only for final reporting. The per-split counts at both the video and cycle level are reported with the results in Section 5.1, since they describe the data on which the controlled benchmarks are computed.

4.2 Experimental Pipelines

Five pipelines are evaluated. They share every stage except cycle source and representation, which is what allows their differences to be attributed to those two factors. Table 4.1 summarises their design and role.

Table 4.1: Experimental pipelines, their cycle source, representation, models, and intended role. Only the first two rows constitute the main controlled comparison; the image pipeline is an extended representation comparison and the last two are supplementary.

Pipeline	Cycle source	Representation	Model(s)	Role
Handmade landmark	Manual segmentation	Landmark + velocity	MLP, 1D-CNN, BiLSTM, CNN-BiLSTM	Upper bound (not deployable)
Autocorr landmark	Autocorr. (`contact_diff`)	Landmark + velocity	MLP, 1D-CNN, BiLSTM, CNN-BiLSTM	Main automatic baseline
Autocorr image	Autocorr. (`contact_diff`)	RGB crop	MobileNetV2–LSTM	Representation comparison
Boundary detector	Learned BiLSTM boundaries	Landmark + velocity	BiLSTM, CNN-BiLSTM	Supplementary
Raw sequence	None (whole video)	Full landmark sequence	GRU, BiLSTM	Supplementary / exploratory

4.2.1 Handmade Landmark Upper Bound

The handmade pipeline uses manually segmented cycles and the landmark representation. It serves as the *upper bound*: it estimates the achievable classification performance when cycle segmentation is reliable, isolating classification difficulty from automatic-detection difficulty. It is explicitly not a deployable baseline.

4.2.2 Autocorr Landmark Baseline

The autocorrelation landmark pipeline is the main *automatic baseline*. It detects cycles from the `contact_diff` signal using the autocorrelation-based detector of Section 3.6.2 and uses the same landmark representation and classifiers as the handmade pipeline, so that the two differ only in cycle source.

4.2.3 Autocorr Image-Based Pipeline

The image-based pipeline reuses the *identical* autocorrelation cycle source as the landmark baseline but represents each phase as a runner-centred RGB crop classified by a MobileNetV2–LSTM model (Section 3.9). Holding the cycle source fixed makes this a controlled comparison of *representation*, separate from any historical image-based development milestone.

4.2.4 Boundary Detector Pipeline

The boundary-detector pipeline replaces rule-based autocorrelation with a learned BiLSTM foot-strike detector (Section 3.6.3); downstream stages are unchanged. It is reported as a supplementary pipeline.

4.2.5 Raw Sequence Pipeline

The raw-sequence pipeline omits cycle cutting entirely and classifies the whole video's pose sequence with a recurrent model (GRU/BiLSTM, Section 3.10.6). Here one sample corresponds to one video, so its evaluation unit differs from the cycle-based pipelines; it is treated as an exploratory, supplementary configuration and is not part of the controlled cycle-level comparison.

4.3 Model Training Configuration

All models are trained on the train partition, selected on the validation partition, and evaluated once on the test partition. The pose-sequence models operate on eight-phase landmark inputs of shape $(B, 8, 78)$, where F=78 features comprise 13 selected joints × 3 coordinates + 3 velocity components (Section 3.7). The layer-by-layer architectures of the four landmark classifiers are shown in Figure 3.7.

The four landmark-sequence models share a common training schedule confirmed from the experiment notebooks: **AdamW** optimiser with learning rate 10^{-3} and weight decay 10^{-4}; **OneCycleLR** scheduler with `pct_start=` 0.2; **BCEWithLogitsLoss** with a class-ratio `pos_weight`; gradient clipping (max norm 1.0); and early stopping with patience 15. Batch size is 16, maximum epochs 80, and random seed 42.

The image model uses eight RGB crops per detected cycle and a MobileNetV2–LSTM architecture [17], [18]. Table 4.2 summarises the per-model configuration. The audited MobileNetV2–LSTM experiment uses learning rate 10^{-3}, batch size 8, 40 epochs, dropout 0.4, two recurrent layers, weight decay 10^{-4}, and random seed 42.

Parameter counts for the four landmark models are taken directly from the exported model-ablation report and are used as a reference for model complexity. The MobileNetV2–LSTM and raw-sequence rows are included for completeness; their parameter counts were not exported. All landmark classifiers were trained under the shared schedule described above; the MobileNetV2–LSTM and raw-sequence models follow separate configurations noted in their respective rows.

Table 4.2: Model configuration confirmed from the experiment notebooks. Parameter counts are taken from the exported model-ablation report. The four landmark-sequence models share a common optimizer/scheduler/loss (see text); per-model architecture details follow Figure 3.7.

Model	Pipeline use	Input	Params	Architecture notes
MLP	Handmade & autocorr landmark	$(B, 8, 78)$	193,025	Flatten → Linear(624→256)+ReLU → Dropout(0.3) → Linear(256→128)+ReLU → Dropout(0.3) → Linear(128→1).
1D-CNN	Handmade & autocorr landmark	$(B, 8, 78)$	40,257	Conv1d(78→64, k=3)+ReLU+BN → Conv1d(64→128, k=3)+ReLU+BN → AdaptiveAvgPool1d(1) → Dropout(0.3) → Linear(128→1).
BiLSTM	Handmade & autocorr landmark	$(B, 8, 78)$	173,441	2-layer bidirectional LSTM (h=64) → concat[h_{fwd}, h_{bwd}] (128-d) → LayerNorm(128) → Dropout(0.3) → Linear(128→1).
CNN-BiLSTM	Handmade & autocorr landmark	$(B, 8, 78)$	82,113	Conv1d(78→64, k=3)+ReLU+BN → 1-layer BiLSTM (h=64) → concat (128-d) → LayerNorm(128) → Dropout(0.3) → Linear(128→1).
MobileNetV2–LSTM	Autocorr image pipeline	8×RGB 224×224	—	MobileNetV2 backbone (frozen or fine-tuned) → 2-layer LSTM (h=256, dropout=0.4) → Linear(→1). lr=10^{-3}; batch=8; epochs=40; seed=42.
GRU / BiLSTM (raw)	Supplementary raw-sequence	Full video landmark seq.	—	Whole-video recurrent model; one sample per video. Treated as supplementary evidence; detailed hyper-parameters not audited.

4.4 Evaluation Metrics

Let TP, FP, FN, and TN denote the counts of true positives, false positives, false negatives, and true negatives for a given class. The following metrics are used.

Accuracy is the fraction of correctly classified samples:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{4.1}$$

Precision and *recall* for a class are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{4.2}$$

measuring, respectively, the proportion of positive predictions that are correct and the proportion of actual positives that are recovered.

The *F1 score* is their harmonic mean:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4.3}$$

The *macro-averaged F1* (macro-F1) is the unweighted mean of the per-class F1 scores over the two classes (correct, false):

$$\text{macro-F1} = \frac{1}{2}\left(\text{F1}_{\text{correct}} + \text{F1}_{\text{false}}\right). \tag{4.4}$$

ROC-AUC is the area under the receiver-operating-characteristic curve, which plots the true-positive rate against the false-positive rate as the decision threshold varies; it summarises ranking quality independently of any single threshold, with 0.5 corresponding to chance [21].

Why macro-F1 is the primary metric. Because the dataset is class-imbalanced at the video level (more false than correct videos, Section 5.1), accuracy can be misleading: a classifier that favours the majority class can attain deceptively high accuracy while failing on the minority class. Macro-F1 weights both classes equally regardless of their frequency, so a model must perform well on *both* correct and false technique to score well [22]. Macro-F1 is therefore reported as the primary metric throughout, with accuracy, per-class F1, and ROC-AUC reported alongside it for completeness. ROC-AUC is particularly useful for separating ranking quality from threshold choice, which is examined explicitly in Section 4.5.7.

4.5 Ablation Study Design

The ablations and extended analyses each vary one factor while holding the remainder of the pipeline fixed, so that the measured effect can be attributed to that factor. Table 4.3 summarises the design; the corresponding results appear in Chapter 5.

Table 4.3: Design of the ablation studies and extended analyses. Each study varies one factor on a fixed pipeline and split.

Study	Varied factor	Pipeline	Held fixed
Model ablation	Classifier (MLP / 1D-CNN / BiLSTM / CNN-BiLSTM)	Handmade & Autocorr	Data, split, representation, threshold
Feature ablation	Feature set (selected, full-33, +velocity, +AP/V, +contact, +all)	Handmade	Model, split, frames
Signal ablation	Detection signal (head-Y, hip-Y, heel-Y, toe-Y, `contact_diff`, `AP_diff`, `V_diff`)	Autocorr	Model, split, representation
Frames-per-cycle	Frames sampled per cycle (8 / 16 / 24 / 32)	Handmade	Model family, split, representation
Representation	Landmark vs. image	Autocorr (shared cycle source)	Cycle source, split
Cycle vs. video	Evaluation unit (cycle vs. video aggregation)	Autocorr (and others)	Model, split
Threshold	Decision threshold (0.5 vs. validation-tuned)	All	Model, split
Pose/cycle quality	Quality factor (visibility, autocorr., missing-ratio, duration, period)	Autocorr	Model, split
Fair-gap robustness	Cycle source under repeated folds and label permutations	Handmade vs. autocorr	MLP, video-level CV, threshold 0.5

4.5.1 Model Ablation

The four classifiers are compared on identical data for both the handmade and autocorr pipelines, holding the representation, split, and decision threshold fixed, to isolate the effect of architecture.

4.5.2 Feature Ablation

On the handmade pipeline, the input feature set is varied — from the compact selected-landmark set through the full landmark set and several derived-feature augmentations — with the classifier and all other stages held fixed, to measure the effect of feature representation.

4.5.3 Signal Ablation

On the autocorr pipeline, the one-dimensional motion signal driving cycle detection is varied across the seven candidates of Table 3.1, holding the classifier fixed. Both detection statistics (coverage, period regularity, rejections) and downstream classification are recorded, since signal choice affects both.

4.5.4 Frames-per-Cycle Ablation

On the handmade pipeline, the number of frames sampled per cycle is varied (8/16/24/32) for the sequence models, holding all else fixed, to test whether temporal resolution beyond eight phases improves classification and at what computational cost.

4.5.5 Representation Ablation

The landmark and image representations are compared on the *same* autocorrelation cycle source and split, so that any difference is attributable to representation rather than to cycle source or protocol. This analysis is kept separate from the historical image-based development milestone, which was recorded under a different protocol and is not a controlled benchmark.

4.5.6 Cycle-Level versus Video-Level Evaluation

Predictions are evaluated both per cycle and aggregated per video (majority vote, mean probability, confidence-weighted vote), holding the model and split fixed, to assess how aggregation affects measured performance. Video-level results are treated as the more faithful estimate of practical performance.

4.5.7 Threshold 0.5 versus Tuned Threshold

For each model, performance at the default 0.5 threshold is compared with performance at a threshold selected on the *validation* set (by maximising validation macro-F1) and applied unchanged to test. Reporting both, together with the change between them, exposes any threshold overfitting rather than concealing it.

4.5.8 Pose and Cycle Quality Analysis

Test cycles from the autocorr pipeline are grouped by pose- and cycle-quality factors — landmark visibility, autocorrelation strength, landmark missing-ratio, cycle duration, and cycle period — and classification performance is compared across groups, to locate the dominant source of error in the automatic pipeline. Because no canonical quality thresholds exist, groups are formed by a median split, and the resulting group sizes are reported with the results so that small groups are interpreted cautiously (Section 5.15).

4.5.9 Fair Gap Robustness Check and Permutation Test

The fixed split gives the primary controlled benchmark. Because the validation and test partitions are small, an additional robustness check is used to test whether the handmade–autocorr gap is specific to that split. Both cycle sources are evaluated with the same MLP classifier, 5-fold video-level cross-validation, the default threshold of 0.5, and macro-F1 as the main metric. Splitting is again performed by video identifier, so all cycles from one video remain within the same fold.

A permutation test is also applied to the autocorr MLP setting. The observed cross-validation macro-F1 is compared with a null distribution obtained by randomly permuting the labels and rerunning the same cross-validation procedure. The test is used as a diagnostic of separability under a simple classifier, not as a replacement for the controlled benchmark. A non-significant result is therefore interpreted conservatively: it indicates that the autocorr-derived features are not robustly separable under that MLP configuration, rather than proving that no useful signal exists.

Chapter 5

Results

This chapter reports the empirical results of the running-technique classification pipelines. Following the experimental protocol of Chapter 4, a strict separation is maintained between (i) *controlled benchmark* results, obtained under the fixed split-by-video protocol with a unified reporting procedure, and (ii) *historical development* results, retained only as method-evolution evidence. All controlled tables use the same fixed train/validation/test split summarised in Section 5.1. A separate fair-gap robustness check then repeats the handmade–autocorr comparison under 5-fold video-level cross-validation to assess whether the fixed-split gap is systematic.

5.1 Dataset Statistics

The dataset is partitioned by video identifier so that all cycles extracted from a given video are confined to a single split, preventing video-specific information from leaking between training and evaluation. Table 5.1 summarises the resulting split.

Table 5.1: Dataset split by video and by detected running cycle. Source: `split_summary.csv`.

Split	**Videos**	**Correct videos**	**False videos**	**Detected cycles**
Train	52	20	32	97
Validation	11	5	6	33
Test	12	5	7	22
Total	**75**	**30**	**45**	**152**

The class balance at the video level is moderately skewed toward the *false* class (45 vs. 30), a distribution that motivates the use of macro-averaged F1 as the primary evaluation metric throughout this chapter, since macro-F1 weights both classes equally and is therefore robust to this imbalance [22]. The small validation and test splits (11 and 12 videos respectively) also caution against over-interpreting single-point differences, a consideration revisited in the discussion of threshold tuning (Section 5.14).

5.2 Handmade Landmark Upper Bound

The handmade landmark pipeline uses manually controlled cycle segmentation and therefore estimates the upper bound of the landmark representation under clean cycle boundaries. It is not treated as a deployment baseline. Table 5.2 reports the four trained architectures.

Table 5.2: Handmade landmark upper-bound results. Best macro-F1 and metrics in bold. Source: `handmade_results.csv`.

Model	**Test acc.**	**Macro-F1**	**F1-correct**	**F1-false**	**ROC-AUC**
MLP	0.8571	0.8545	0.8293	0.8798	0.9444
1D-CNN	0.8929	0.8772	0.8889	0.8649	0.9444
BiLSTM	0.8750	0.8623	0.8889	0.8357	0.9556
CNN-BiLSTM	**0.9107**	**0.8991**	**0.9333**	**0.8649**	**0.9556**

The handmade setting establishes that the landmark representation is learnable when cycle segmentation is reliable, with all four architectures exceeding 0.85 macro-F1 and ROC-AUC above 0.94. This directly answers the first research question: when running cycles are correctly segmented, the markerless pose representation carries sufficient information for a deep model to discriminate correct from false technique. The narrow spread across architectures (macro-F1 0.8545–0.8991) indicates that, given clean cycles, classification is not strongly architecture-limited.

5.3 Autocorr Landmark Baseline

The autocorrelation landmark pipeline is the main automatic baseline. It uses the `contact_diff` motion signal, autocorrelation period estimation, foot-strike-to-foot-strike cycle cutting, no cycle margin, selected landmark features, and velocity features. Table 5.3 reports the controlled fixed-split results.

Table 5.3: Autocorr landmark baseline results. Best macro-F1 in bold. Source: `autocorr_results.csv`.

Model	Test acc.	Macro-F1	F1-correct	F1-false	ROC-AUC
MLP	0.5455	0.5417	0.5000	0.5833	0.6068
1D-CNN	0.5000	0.3425	0.6850	0.0000	0.5684
BiLSTM	0.4545	0.3714	0.7429	0.0000	0.5342
CNN-BiLSTM	**0.5909**	**0.5901**	0.5333	0.6468	**0.6370**

Although the CNN-BiLSTM has the best fixed-threshold macro-F1 in the raw autocorr model table, the thesis baseline uses the MLP row in later summary tables because the threshold analysis in Section 5.14 shows it is more stable after validation-based threshold selection. This distinction is important: the automatic baseline is evaluated not only by its peak fixed-threshold score but also by its robustness to the complete reporting protocol. Compared with the handmade upper bound, the autocorr landmark baseline drops substantially. The best fixed-threshold automatic result reaches only 0.5901 macro-F1, and the robust reported MLP baseline reaches 0.5417 macro-F1. Some sequential models collapse on the *false* class so dominantly that F1-false reaches 0.0000. This validation-to-test collapse, combined with ROC-AUC values close to 0.5, indicates that the limiting factor in the automatic pipeline is not classifier capacity but the quality and consistency of the automatically segmented cycles. The contrast with Section 5.2 is quantified directly in Section 5.4.

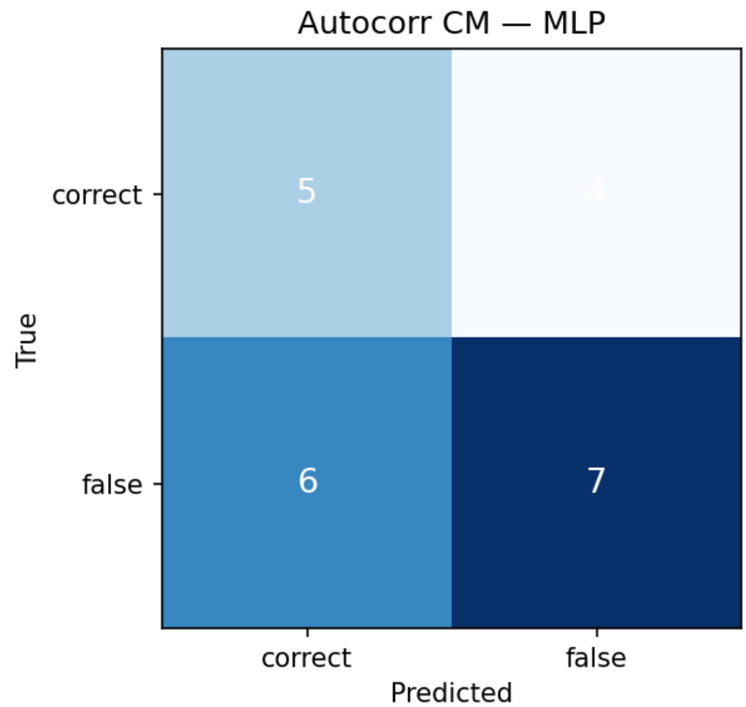


Figure 5.1: Confusion matrix for the autocorrelation-based landmark pipeline on the controlled test split.

5.4 Handmade versus Autocorr Comparison

Table 5.4 gives the main controlled comparison between clean manual cycles and automatic autocorrelation cycles. The rows use the same landmark representation family, so the comparison isolates the effect of cycle source.

Replacing handmade cycles with automatic cycles reduces accuracy by 0.3652 (from 0.9107 to 0.5455) and macro-F1 by 0.3574 (from 0.8991 to 0.5417). Because both pipelines share the same landmark representation, classifier family, split, and reporting protocol, this gap is attributable primarily to the cycle-segmentation stage rather than to the classifier. This is the central empirical finding of the thesis: the markerless representation is *learnable* (Section 5.2) but the automatic *segmentation* of in-the-wild running cycles is the dominant bottleneck. The detection-stage statistics also show why: the `contact_diff` pipeline retains

Table 5.4: Main comparison between handmade upper bound and autocorr automatic landmark baseline. Best controlled result in bold. Source: `main_pipeline_comparison.csv`.

Pipeline	Cycle source	Test acc.	Macro-F1	F1-correct	F1-false	ROC-AUC
Handmade landmark upper bound	Manual	**0.9107**	**0.8991**	**0.9333**	**0.8649**	**0.9556**
Autocorr landmark baseline	Automatic	0.5455	0.5417	0.5000	0.5833	0.6068

only 151 of its raw cycles and fails to detect usable cycles in a substantial number of videos (Section 5.8). Section 5.5 tests the same interpretation under repeated video-level folds.

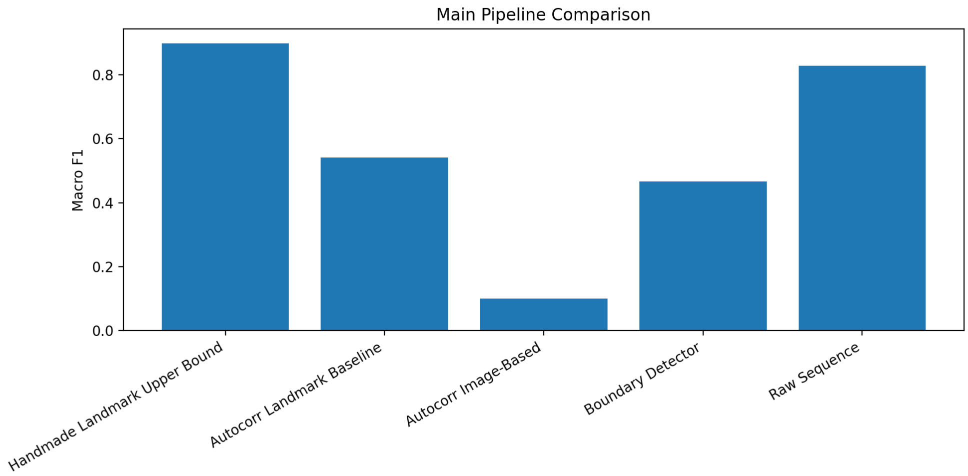


Figure 5.2: Main pipeline comparison between the handmade landmark upper bound, the autocorr landmark baseline, and supplementary pipelines. Source: `main_pipeline_comparison.png`.

5.5 Fair Gap Robustness Check

The fixed-split comparison in Section 5.4 gives the primary controlled benchmark, but the validation and test sets are small. To test whether the handmade–autocorr gap was merely a difficult split artefact, an additional fair-gap robustness check was conducted. Both cycle sources were evaluated using the same MLP classifier, 5-fold video-level cross-validation, a fixed threshold of 0.5, and macro-F1 as the main metric.

Table 5.5 confirms the same conclusion as the fixed-split benchmark. Handmade cycles remain stable under repeated video-level folds, achieving 0.860 ± 0.016 macro-F1. Autocorr cycles remain substantially weaker, achieving 0.615 ± 0.081 macro-F1. The resulting fair gap of 0.245 macro-F1 is smaller than the single fixed-split gap of 0.3574, but it remains large enough to show that the performance loss is systematic rather than only the result of one difficult split.

Table 5.5: Fair-gap robustness check under 5-fold video-level cross-validation using the same MLP classifier and threshold 0.5. Source: `outputs/autocorr/fair_gap_and_mlp_perm.csv`.

Cycle source	Cycles	Videos	CV macro-F1	OOF macro-F1, 95% CI
Handmade cycles	370	369	0.860 ± 0.016	0.860 [0.822, 0.897]
Autocorr cycles	151	55	0.615 ± 0.081	0.655 [0.580, 0.728]
Fair gap (handmade − autocorr)			**0.245**	–

A permutation test was then applied to the autocorr MLP setting to assess whether the observed autocorr features were robustly separable under the same simple classifier. With 100 random label permutations, the observed CV macro-F1 was 0.5206, while the null distribution had a mean of 0.468 and a standard deviation of 0.074, giving $p = 0.2475$. This does not establish statistically significant separation from the permutation baseline. The result should be read conservatively: autocorr-derived cycles contain some discriminative signal, as also reflected by the CV comparison, but the signal is noisy and not robustly separable under the tested MLP configuration.

Table 5.6: Permutation test for the autocorr MLP setting. Source: `outputs/autocorr/fair_gap_and_mlp_perm.json`.

Quantity	Value
Observed autocorr CV macro-F1	0.5206
Null mean macro-F1	0.4680
Null standard deviation	0.0740
Number of permutations	100
Permutation p-value	0.2475

5.6 Model Ablation Results

The model ablation compares MLP, 1D-CNN, BiLSTM, and CNN-BiLSTM architectures under the handmade and autocorr pipelines. Table 5.7 shows the macro-F1 and accuracy for each setting.

The architecture effect is pipeline-dependent. In the handmade setting, the hybrid CNN-BiLSTM performs best, suggesting that local pose patterns and temporal ordering are both useful when the eight phases are correctly aligned. In the automatic setting, model ranking becomes unstable: at a 0.5 threshold the CNN-BiLSTM is best (macro-F1 0.5901), whereas the simpler MLP wins once thresholds are tuned (Section 5.14). The absolute values remain low for every automatic-pipeline model, reinforcing that architecture choice cannot recover the information lost to imperfect segmentation.

Table 5.7: Model ablation on handmade and autocorr pipelines. Best macro-F1 within each pipeline in bold. Source: `model_ablation.csv`.

Pipeline	Model	Test acc.	Macro-F1	ROC-AUC
Handmade	MLP	0.8571	0.8545	0.9444
	1D-CNN	0.8929	0.8772	0.9444
	BiLSTM	0.8750	0.8623	**0.9556**
	CNN-BiLSTM	**0.9107**	**0.8991**	**0.9556**
Autocorr	MLP	0.5455	0.5417	0.6068
	1D-CNN	0.5000	0.3425	0.5684
	BiLSTM	0.4545	0.3714	0.5342
	CNN-BiLSTM	**0.5909**	**0.5901**	**0.6370**

5.7 Feature Ablation Results

The feature ablation fixes the classifier and varies only the input feature set, on the handmade pipeline. Table 5.8 reports the resulting performance.

Table 5.8: Feature ablation on handmade cycles. Best macro-F1 in bold. Source: `feature_ablation.csv`.

Feature set	Test acc.	Macro-F1	F1-false	ROC-AUC
Selected 13 landmarks	0.9107	**0.8991**	0.8649	**0.9556**
All 33 landmarks	0.8929	0.8961	**0.9189**	0.9389
Lower-body only	0.8929	0.8961	**0.9189**	0.9222
Selected 13 + velocity	0.9107	**0.8991**	0.8649	**0.9556**

The selected 13-landmark representation matches the best macro-F1 while remaining lower-dimensional than the full 33-landmark representation. The lower-body-only and all-landmark variants improve F1-false but do not improve macro-F1 overall, suggesting that additional coordinates can alter class balance without yielding a more robust classifier. The selected-landmark configurations cluster tightly around 0.90 macro-F1, supporting the use of the compact 13-landmark representation as the pipeline default for the automatic experiments, where added input dimensionality offers little benefit relative to its sensitivity to landmark noise.

5.8 Signal Ablation Results

The signal ablation compares the seven candidate motion signals used for cycle detection, holding the classifier family fixed. Table 5.9 reports detection coverage and classification performance.

The `hip_y` signal achieves the highest classification macro-F1 in this ablation, while `contact_diff` is retained as the main baseline signal because it reflects foot-contact dy-

Table 5.9: Cycle-detection signal ablation. Best macro-F1 and best retained cycle count are bolded separately. Source: `signal_ablation.csv`.

Signal	Test acc.	Macro-F1	Videos OK	Cycles retained	Cycles rejected
head_y	0.6818	0.6703	**72**	**204**	13
hip_y	**0.7333**	**0.7321**	67	190	15
heel_y	0.6667	0.6623	35	49	36
toe_y	0.5000	0.5000	35	52	33
contact_diff	0.5909	0.5901	50	151	42
AP_diff	0.2500	0.2000	32	33	39
V_diff	0.6364	0.6022	48	125	34

namics and produces more semantically aligned cycle boundaries. The decision is not based on accuracy alone. In this thesis, the baseline signal must support foot-strike-to-foot-strike sampling, eight-phase interpretation, and stable comparison to the image pipeline. The `contact_diff` signal therefore produces the most temporally coherent foot-strike-to-foot-strike boundaries, consistent with periodic gait-cycle structure in human motion **cutting1978gaitperiodicity**, [23], which is why it is adopted as the baseline cycle-detection signal even though its classification macro-F1 (0.5901) is not the single highest. The vertical signals (`head_y`, `hip_y`) detect more videos but admit many ill-formed cycles, illustrating the trade-off between detection coverage and boundary precision that underlies the segmentation bottleneck identified in Section 5.4.

5.9 Frames-per-Cycle Ablation Results

The frame ablation tests whether increasing temporal resolution beyond the eight-phase representation improves performance on handmade cycles. Table 5.10 summarises the best model at each frame count.

Table 5.10: Frames-per-cycle ablation on handmade cycles. Best macro-F1 in bold. Source: `frames_ablation.csv`.

Frames per cycle	Best model	Test acc.	Macro-F1	ROC-AUC
8	CNN-BiLSTM	0.9107	**0.8991**	**0.9556**
16	BiLSTM	0.8750	0.8623	**0.9556**
24	CNN-BiLSTM	0.8393	0.8347	0.9389
32	CNN-BiLSTM	0.8571	0.8545	0.9444

The eight-phase representation gives the best macro-F1 despite being the shortest sequence. Sixteen frames remain highly competitive (BiLSTM macro-F1 0.8623 at the lowest cost), and increasing frame count yields no monotonic improvement. This supports the eight-phase sampling choice as an efficient representation of the running cycle, with additional temporal resolution providing diminishing returns on this dataset.

5.10 Autocorr Version History and Method Evolution

This section documents the evolution of the autocorrelation pipeline. The values in Table 5.11 are *historical* validation figures drawn from development notebooks; they are presented as method-evolution evidence only and must not be read as controlled benchmark results. Where a metric was never recorded under the controlled protocol it is marked `[MISSING RESULT]`.

Table 5.11: Autocorr version history (historical development evidence — not a controlled benchmark). Source: `autocorr_version_comparison.csv`.

Version	Signal	Input	Model	Val. acc. (%)
`autocorr_fixed_val_acc_69_6`	head-Y oscillation	landmarks	CNN+LSTM	69.6
`autocorr_fixed_val_acc_72_9`	`contact_diff`	landmarks	CNN+LSTM	72.9
`autocorr_fixed_biLSTM`	`contact_diff`	landmarks + vel.	BiLSTM	65.0
`autocorr_fixed_val_acc_76_5`	`contact_diff`	RGB crop 224^2	MobileNetV2+LSTM	76.5
`02_baseline_landmark`	`contact_diff`	13 lm + vel.	best of 4	†

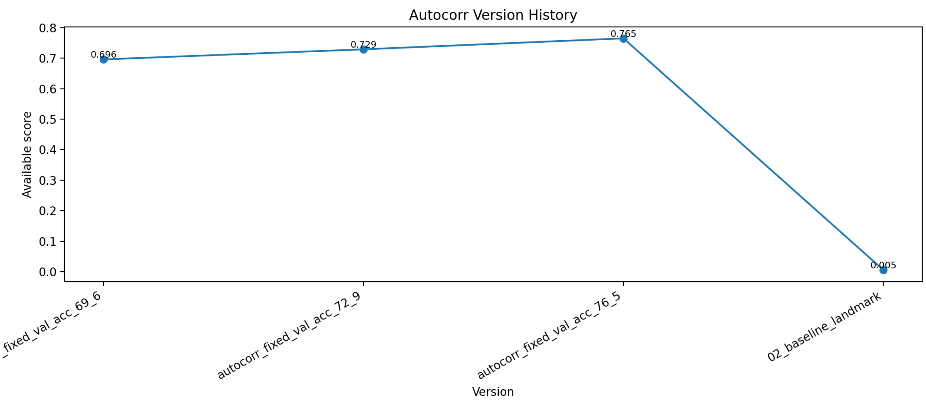


Figure 5.3: Historical autocorr method evolution. This figure is development evidence only; Source: `autocorr_version_history.png`.

† The `02_baseline_landmark` row is the only controlled entry; under the fixed-split protocol it records a test accuracy of 0.5455 and a test macro-F1 of 0.5417 (Section 5.3), in [0,1] rather than percent. Its historical validation accuracy was not recorded (`[MISSING RESULT]`).

The development trace shows three substantive changes: replacing the noisy head-Y oscillation signal with `contact_diff` (historical validation 69.6% → 72.9%), adopting autocorrelation period estimation with foot-strike-to-foot-strike alignment, and an exploratory shift to an image-based representation (`autocorr_fixed_val_acc_76_5`). The 76.5% figure is the most technically complex historical milestone, but it is a development validation accuracy on an RGB-crop representation and was not re-confirmed under the controlled protocol; it is therefore retained as motivation for the organised image-based rerun (Section 5.11) rather than as evidence that image-based modelling outperforms the landmark baseline.

5.11 Autocorr Image-Based Results

The image-based pipeline is the *controlled* rerun of the historical image idea: it reuses the exact `contact_diff`/autocorrelation cycle source of the landmark baseline, but represents each phase as a runner-centred RGB crop processed by a MobileNetV2+LSTM model [15], [17], [18]. Table 5.12 reports the controlled result.

Table 5.12: Autocorr image-based pipeline (controlled rerun). Source: `image_results.csv`.

Input	Model	Test acc.	Macro-F1	F1-false	ROC-AUC
RGB crop	MobileNetV2+LSTM	0.1111	0.1000	0.2000	0.0592

Under the controlled protocol the image-based pipeline reaches a test macro-F1 of only 0.1000, with F1-correct of 0.0000 and a ROC-AUC of 0.0592, i.e. far below chance. The model effectively fails to learn a usable decision boundary on this small dataset. Crucially, this controlled result does *not* reproduce the historical 76.5% validation figure of Section 5.10, which underscores why that figure cannot serve as a benchmark. The collapse is consistent with the known fragilities of the image representation on limited in-the-wild data: bounding-box and crop instability, domain shift, and overfitting of a high-capacity visual backbone [18]. This indicates that the current image-based implementation is not yet stable enough to replace the landmark baseline, not that image-based modelling is invalid in principle.

5.12 Landmark-Based versus Image-Based Representation

Table 5.13 compares the two representations on the same automatic cycle source, at the cycle (sample) level.

Table 5.13: Landmark versus image representation (controlled), identical cycle source. Source: `representation_comparison.csv`.

Representation	Model	Test acc.	Macro-F1	ROC-AUC	Test samples
Landmark	MLP	**0.5455**	**0.5417**	0.6068	22
Image	MobileNetV2+LSTM	0.1111	0.1000	0.0592	27

On a matched cycle source, the landmark representation clearly outperforms the image representation (macro-F1 0.5417 versus 0.1000).The same ordering holds at the video level, where the landmark pipeline reaches 0.6250 accuracy / 0.6190 macro-F1 under majority voting against 0.2857 / 0.2222 for the image pipeline (`representation_video_level_comparison.csv`). For this dataset and implementation, the sparse landmark sequence is

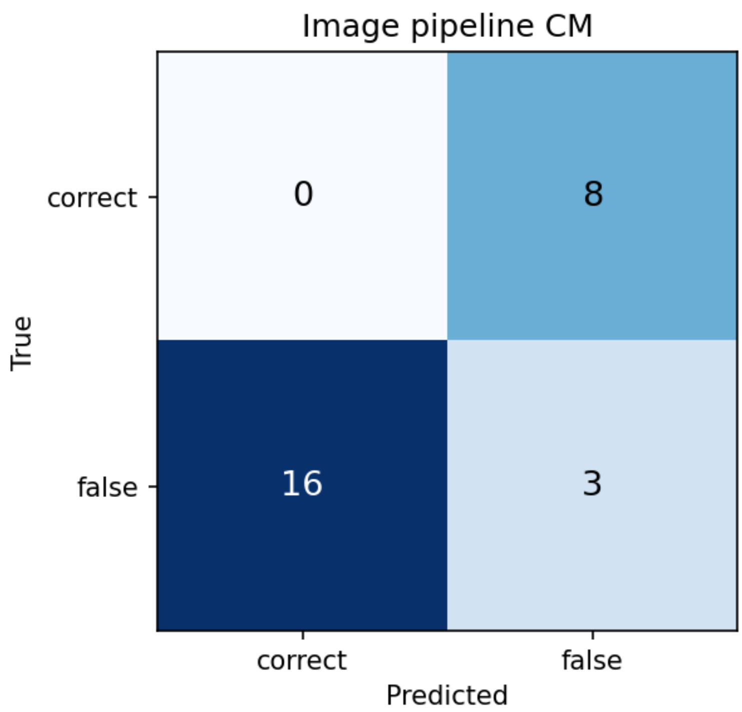


Figure 5.4: Confusion matrix for the image-based pipeline on the controlled test split.

the more reliable representation; the RGB-crop pipeline does not add usable discriminative information and is markedly less stable. This addresses the representation research question while remaining within the controlled protocol, separately from the historical 76.5% milestone.

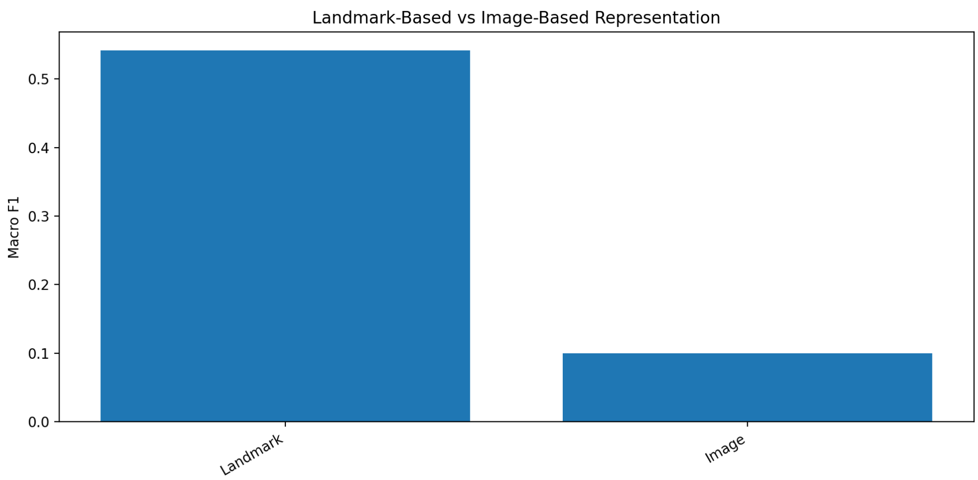


Figure 5.5: Controlled landmark-versus-image representation comparison under the automatic autocorr cycle source. Source: `landmark_vs_image_comparison.png`.

5.13 Cycle-Level versus Video-Level Evaluation

Because each video may yield several cycles, predictions can be reported per cycle or aggregated per video. Table 5.14 compares the two for the automatic landmark baseline (the main pipeline) and lists the supplementary pipelines for reference.

Table 5.14: Cycle-level versus video-level evaluation. Video-level uses majority vote. Sources: `cycle_level_metrics.csv`, `video_level_metrics.csv`.

Pipeline	Level	Acc.	Macro-F1	F1-false	N
Autocorr (baseline)	Cycle	0.5238	0.5139	0.5833	21
	Video (majority)	**0.5714**	**0.5333**	0.6667	7
Image	Cycle	0.1111	0.1000	0.2000	27
Image	Video (majority)	0.2857	0.2222	0.4444	7
Boundary Detector	Cycle	0.6000	0.5833	0.5000	5
Boundary Detector	Video (majority)	0.6667	0.6667	0.6667	3
Raw sequence	Cycle/Video	0.8182	0.8036	0.8571	11

For the automatic landmark baseline, aggregating cycle predictions to the video level by majority vote improves both accuracy (0.5238 → 0.5714) and macro-F1 (0.5139 → 0.5333). The same direction appears for the image and boundary-detector pipelines. This indicates that per-cycle predictions contain noise that partially cancels under aggregation, so video-level reporting gives a fairer estimate of practical performance. The very small video counts (N = 3–7) mean these gains, while consistent in direction, rest on few samples and should be treated as indicative.

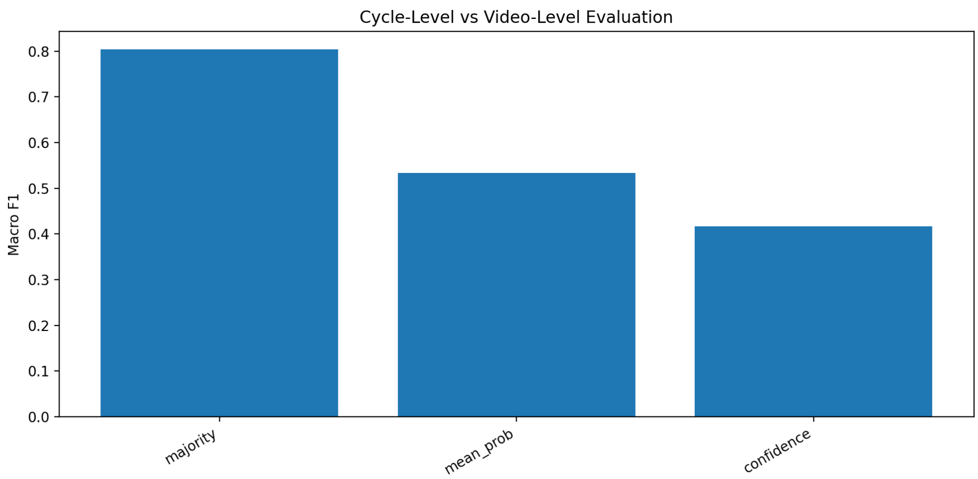


Figure 5.6: Cycle-level versus video-level evaluation. Video-level predictions aggregate cycle predictions by majority voting. Source: `cycle_vs_video_comparison.png`.

5.14 Threshold 0.5 versus Tuned Threshold

Decision thresholds tuned on validation can either improve or overfit test performance. Table 5.15 reports, for each model, the macro-F1 at the default 0.5 threshold and at the validation-selected threshold, together with the change and an overfit flag.

Table 5.15: Default threshold versus validation-tuned threshold. Thresholds are selected on validation and applied to test. Source: `threshold_05_vs_tuned.csv`.

Pipeline	Model	Macro-F1@0.5	Best thr.	Tuned macro-F1	Δ	Overfit
Handmade	1D-CNN	0.8961	0.34	0.8772	−0.0189	True
	BiLSTM	0.8772	0.78	0.8772	0.0000	False
	CNN-BiLSTM	0.8772	0.40	**0.8991**	+0.0219	False
	MLP	0.8250	0.16	0.8545	+0.0295	False
Autocorr	MLP	0.5417	0.34	**0.5417**	0.0000	False
	1D-CNN	0.3425	0.50	0.3425	0.0000	False
	BiLSTM	0.3714	0.54	0.2903	−0.0811	True
	CNN-BiLSTM	0.5901	0.18	0.4359	−0.1542	True
Image	all	0.1000	[MISSING RESULT]	[MISSING RESULT]	[MISSING RESULT]	False

On the clean handmade pipeline, validation-tuned thresholds mostly help or are neutral (CNN-BiLSTM improves to 0.8991). On the automatic pipeline the picture reverses: tuning leaves the MLP unchanged but severely degrades the sequential models, with the CNN-BiLSTM dropping by 0.1542 macro-F1 and being flagged as threshold-overfit. This explains the apparent disagreement with the model ablation (Section 5.6): the CNN-BiLSTM is best at the 0.5 threshold but unstable once a validation-selected threshold is applied, whereas the MLP is the robust choice for the reported baseline. The image pipeline exposes only an aggregate row with no probability-based sweep, so its tuned entries are `[MISSING RESULT]`. All reported baseline metrics state explicitly whether a 0.5 or tuned threshold was used.

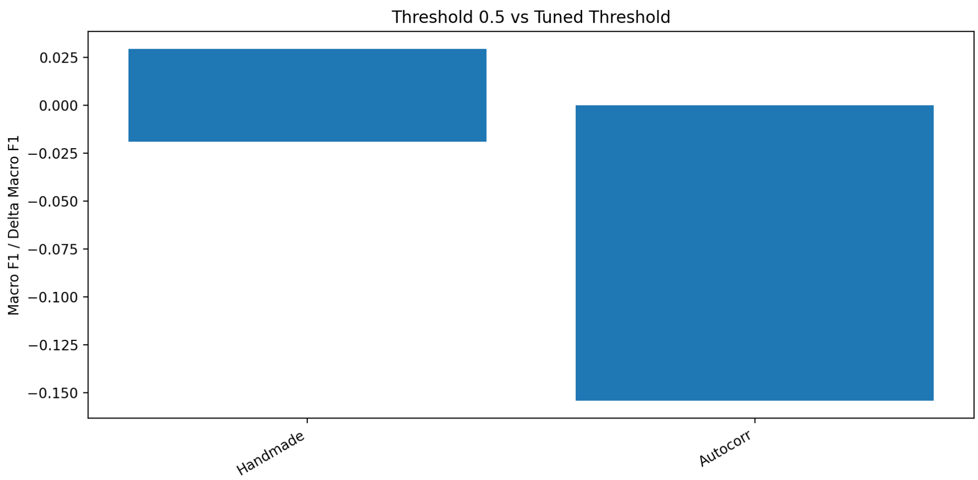


Figure 5.7: Comparison between the default 0.5 decision threshold and validation-tuned thresholds. Source: `threshold_comparison.png`.

5.15 Pose and Cycle Quality Analysis

To locate the source of automatic-pipeline errors, test cycles are grouped by pose- and cycle-quality factors (median split) and accuracy/macro-F1 compared across groups. Table 5.16 reports representative factors; within each factor the higher-accuracy group is bold.

Table 5.16: Pose and cycle quality analysis (Autocorr baseline). Median-split groups. Source: `pose_cycle_quality_metrics.csv`.

Quality factor	Group	n	Acc.	Macro-F1
Heel visibility	low (≤ 0.91)	11	0.4545	0.4107
	high (> 0.91)	11	**0.6364**	0.5417
Autocorr correlation	low (≤ 0.59)	18	0.5000	0.4582
	high (> 0.59)	4	**0.7500**	0.4286
Missing-ratio proxy	low (≤ 0.05)	11	**0.6364**	0.5417
	high (> 0.05)	11	0.4545	0.4107
Cycle duration	low (≤ 0.88)	16	0.5000	0.4921
	high (> 0.88)	6	**0.6667**	**0.6667**
Cycle period	low (≤ 0.77)	12	**0.5833**	0.5556
	high (> 0.77)	10	0.5000	0.3333

The accuracy trends are consistent with the segmentation-bottleneck interpretation: cycles with higher heel visibility (0.6364 vs. 0.4545), lower landmark missing-ratio (0.6364 vs. 0.4545), and stronger autocorrelation (0.7500 vs. 0.5000) are classified more accurately. In other words, errors concentrate where pose estimation is unreliable or the periodic signal is weak, rather than being uniformly distributed. Several high-quality groups are very small (n = 4–6), so the magnitudes are indicative; the direction, however, points clearly at upstream pose and cycle quality as the dominant error source for the automatic pipeline. A complementary breakdown of failed videos (`failed_video_quality_summary.csv`) attributes the majority of failures to the cycle-detection stage.

5.16 Boundary Detector and Raw Sequence Results

Two exploratory alternatives to rule-based cycle detection are reported here as supplementary evidence. The learned Boundary Detector replaces autocorrelation with a BiLSTM foot-strike detector; the raw-sequence model classifies the whole pose sequence without explicit cycle cutting (one sample per video). Table 5.17 summarises their best configurations.

The Boundary Detector (best macro-F1 0.4667) does not improve on the autocorrelation baseline and remains limited by the same scarce, noisy cycle boundaries. The raw-sequence BiLSTM reports a high macro-F1 (0.8286), but this figure is computed over

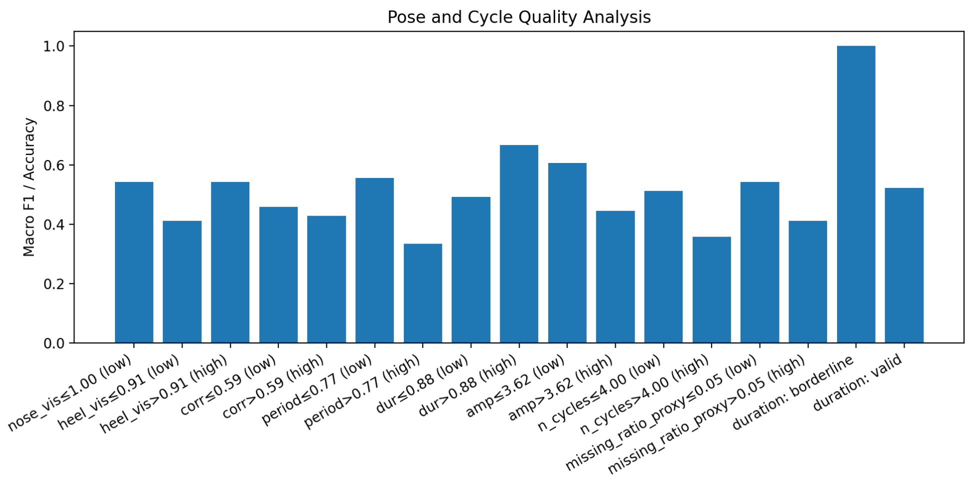


Figure 5.8: Pose and cycle quality summary for the automatic landmark pipeline. Source: `quality_analysis_summary.png`.

Table 5.17: Boundary Detector and raw-sequence pipelines (Supplementary results). Sources: `main_pipeline_comparison.csv`, `bd_results.csv`, and `raw_sequence_results.csv`.

Pipeline	Best config.	Test acc.	Macro-F1	F1-false	ROC-AUC
Boundary Detector	BiLSTM	0.5000	0.4667	0.3333	`[MISSING RESULT]`
Raw sequence	BiLSTM, seq= 16, selected	0.8333	0.8286	0.8571	`[MISSING RESULT]`

only 12 video-level test samples and uses a different evaluation unit (one sequence per video) than the cycle-level benchmarks; it is therefore reported as an exploratory direction rather than as evidence that cycle cutting is unnecessary. ROC-AUC was not stored for either pipeline (`[MISSING RESULT]`). Both are kept out of the main controlled comparison.

5.17 Summary of Main Findings

Table 5.18 consolidates the controlled pipelines alongside the supplementary ones. The best *controlled benchmark* is bold. The image-based row is shown without bold emphasis, in line with its exploratory status, and the historical 76.5% milestone of Section 5.10 is deliberately excluded from this benchmark table.

Table 5.18: Summary of pipeline results. The best controlled benchmark is bolded. Source: `main_pipeline_comparison.csv`.

Pipeline	Input	Model	Acc.	Macro-F1	F1-false	ROC-AUC
Controlled benchmark						
Handmade upper bound	13 lm + vel.	CNN-BiLSTM	**0.9107**	**0.8991**	0.8649	0.9556
Autocorr landmark baseline	landmark feats.	MLP	0.5455	0.5417	0.5833	0.6068
Autocorr image-based	RGB crop	MobileNetV2+LSTM	0.1111	0.1000	0.2000	0.0592
Supplementary; small N, not controlled benchmark						
Boundary Detector	landmark seq.	BiLSTM	0.5000	0.4667	0.3333	`[MISSING RESULT]`
Raw sequence	selected, seq= 16	BiLSTM	0.8333	0.8286	0.8571	`[MISSING RESULT]`

The results give a coherent answer to the thesis research questions. First, the markerless pose representation is sufficient to classify correct versus false running technique when cycles are clean: the handmade upper bound reaches 0.8991 macro-F1. Second, automatic cycle detection is the dominant bottleneck: the matched automatic baseline falls to 0.5417 macro-F1, a fixed-split gap of 0.3574. A separate 5-fold video-level CV check confirms that this is not only a single-split artefact: handmade cycles reach 0.860 ± 0.016 macro-F1, while autocorr cycles reach 0.615 ± 0.081, leaving a fair gap of 0.245. Together, the ablations and robustness check attribute the loss to segmentation and pose quality rather than to the classifier or temporal resolution. Third, within the controlled protocol the compact landmark representation is more reliable than the RGB-crop image representation, which collapses on this small dataset. The supplementary boundary-detector and raw-sequence pipelines do not overturn these conclusions. Taken together, the evidence supports investing future work in robust cycle segmentation and pose quality rather than in larger classifiers or richer image representations. Consistent with this scope, the results do not claim real-world deployment readiness.

Chapter 6

Discussion

This chapter interprets the empirical results of Chapter 5. The guiding question is not only *how well* the pipelines perform, but *where* performance is gained and lost, and *what that implies* for the dataset and protocol used in this work.

6.1 Interpretation of the Main Findings

The central pattern across all experiments is a decoupling between *representation learnability* and *automatic segmentation reliability*. When running cycles are correctly segmented, the markerless pose representation is clearly sufficient: the handmade upper bound separates correct from false technique at a macro-F1 near 0.90 (Section 5.2) . When the same representation and classifiers are driven by automatically detected cycles, performance falls to roughly chance (Section 5.3) . The natural reading is that the classification *model* is not the binding constraint; the binding constraint is the upstream pipeline that converts raw, uncontrolled video into clean, comparable cycle samples.

This conclusion is supported at two levels. The fixed-split benchmark shows a drop from 0.8991 macro-F1 for handmade cycles to 0.5417 for the autocorr baseline. The fair-gap robustness check then repeats the comparison with the same MLP classifier under 5-fold video-level cross-validation: handmade cycles reach 0.860 ± 0.016 macro-F1, while autocorr cycles reach 0.615 ± 0.081, leaving a fair gap of 0.245. Thus, the loss is not only a single-split artefact; it remains visible when the comparison is repeated across video-level folds.

This interpretation reframes the contribution of the thesis. The work does not claim a high-accuracy automatic classifier. Instead, it provides evidence, through a matched upper-bound/baseline design and a series of ablations, that the difficulty of in-the-wild running-technique analysis lies predominantly in robust cycle detection and pose quality rather than in the choice of classifier or representation depth. Every subsequent section in this chapter can be read as locating one facet of that bottleneck.

6.2 Effect of Cycle Source

The comparison between the handmade and autocorrelation (autocorr) pipelines (Section 5.4) is the most informative single contrast in the study. Because the two pipelines share the landmark representation, classifier family, split, and reporting protocol, the large gap between them isolates the effect of cycle segmentation almost cleanly: the manual and automatic pipelines differ chiefly in *how cycle boundaries are obtained*. The fair-gap robustness check strengthens this interpretation: under the same MLP classifier and 5-fold video-level cross-validation, handmade cycles achieve 0.860 ± 0.016 macro-F1, whereas autocorr cycles achieve 0.615 ± 0.081. The 0.245 macro-F1 fair gap shows that the fixed-split drop is not simply due to one unlucky train/test partition.

Two mechanisms plausibly explain the gap. First, automatic detection discards a substantial fraction of the data: many videos yield no usable cycle at all, and of those that do, a large share of raw cycles are rejected by the duration/quality filters (Section 5.8). The surviving cycles are therefore both fewer and biased toward easier cases. Second, even retained cycles may have imprecise foot-strike-to-foot-strike boundaries, so that the eight-phase samples are not temporally aligned across examples. A classifier that performed well on manually aligned phases then faces a distribution of misaligned, partially mis-segmented inputs.

The validation-to-test collapse of the sequential models under automatic cycles (Section 5.3) is consistent with this misalignment: the models latch onto regularities that do not transfer once boundaries vary. The practical message is that improving the automatic pipeline should target detection coverage and boundary precision before any change to the classifier.

The permutation test further tempers the interpretation of the autocorr signal. The observed autocorr MLP CV macro-F1 was 0.5206, compared with a permutation null mean of 0.468 and $p = 0.2475$. This does not prove that the automatic features contain no information; rather, it suggests that the information is weak and unstable under a simple MLP configuration. This is consistent with a pipeline whose errors originate upstream in cycle boundary quality.

6.3 Effect of Representation: Landmark versus Image

On a matched automatic cycle source, the landmark representation clearly outperformed the RGB-crop image representation (Section 5.12), and the image pipeline collapsed below chance in the controlled rerun (Section 5.11). At first sight this is surprising, since RGB crops carry strictly more raw information (silhouette, limb contour, clothing-relative motion) than sparse landmarks.

The likely explanation is not that this information is useless, but that the image pipeline cannot exploit it reliably under the present conditions: a small training set, a high-capacity visual backbone prone to overfitting, and crop instability driven by the same noisy pose estimates that bound the boundary quality. In effect, the image pipeline inherits the segmentation bottleneck and adds a second source of variance — the runner-centred crop — on top of it.

The landmark representation, by contrast, acts as a compact inductive bias: by reducing each frame to a fixed set of joint coordinates, it discards appearance variation that would otherwise have to be learned away from limited data. For this dataset, that compression is an advantage rather than a loss. The appropriate conclusion is conditional: image-based modelling is not invalidated in principle, but the current implementation is not yet stable enough to compete with the landmark baseline. A fair re-test would require more data and more robust cropping.

This is also the right place to interpret the historical `autocorr_fixed` (val_acc 76.5) milestone (Section 5.10) . It represents the most technically ambitious step in the project's development — the move from landmark-only sequences to an image-based MobileNetV2–LSTM pipeline — and it motivated the organised image-based rerun. However, its 76.5% figure is a historical validation accuracy from a development notebook, recorded under a different protocol and never re-confirmed on the fixed test split. The controlled rerun did not reproduce it. The milestone is therefore valuable as evidence of methodological exploration and as the reason the image direction was worth testing, but it cannot serve as a benchmark and must not be compared directly against the controlled landmark baseline.

6.4 Effect of Model Architecture

The model ablation (Section 5.6) shows that architecture matters far less than data quality. On the clean handmade data, the CNN-BiLSTM achieved the strongest macro-F1, while the 1D-CNN, BiLSTM, and MLP remained close enough to show that the spread across architectures was narrow. This indicates that, given well-segmented cycles, the discriminative signal is accessible to several standard architectures and does not depend solely on deep temporal machinery.

On the automatic pipeline the more expressive sequential models behaved unstably. The 1D-CNN and BiLSTM both produced zero F1 for the false class at their operating thresholds, while the CNN-BiLSTM looked best at the default 0.5 threshold but became unstable under validation-tuned thresholding. High-capacity sequence models are precisely those most able to overfit the spurious regularities introduced by inconsistent segmentation, so their poor generalisation is symptomatic of the upstream noise rather than of an architectural deficiency. The robustness of the simple MLP as the reported automatic baseline follows from the same logic: with noisy, scarce inputs, a lower-variance model is preferable. This argues against reaching for larger classifiers as a remedy for the automatic pipeline's weakness.

6.5 Effect of Signal Choice

The signal ablation (Section 5.8) exposed a tension between two notions of a "good" detection signal. The vertical-oscillation signals (`head_y`, `hip_y`) detected cycles in more videos and, for `hip_y`, even produced the highest downstream classification macro-F1. Yet they admitted many ill-formed cycles, reflected in weak mean autocorrelation and a large number of duration-based rejections.

The `contact_diff` signal showed a different profile: fewer detected videos, but cycle boundaries that were more directly tied to foot-contact events and therefore more suitable for foot-strike-to-foot-strike sampling. Selecting `contact_diff` as the baseline signal therefore reflects a deliberate preference for boundary *precision* over detection *coverage*. This is the defensible choice for a pipeline whose downstream representation is an eight-phase sampling that assumes accurate foot-strike-to-foot-strike boundaries: a cleanly segmented minority is more useful than a larger but temporally inconsistent set.

The fact that `hip_y` scored a higher macro-F1 is interpreted cautiously, given the small test set and the risk that more permissive detection coincided with easier videos rather than with genuinely better segmentation. The broader lesson is that signal selection for in-the-wild data must be judged on segmentation quality, not only on a single end-to-end accuracy number.

6.6 Effect of Feature Representation

The feature ablation (Section 5.7) found that the selected 13-landmark representation and the selected-landmark-plus-velocity variant matched the best handmade macro-F1, while the full 33-landmark and lower-body-only variants remained close but did not improve the overall macro-F1. The conservative interpretation is that additional landmarks may carry complementary cues for one class, but they did not produce a more robust overall result on this split.

The near-equivalence of the selected-landmark configurations supports using the compact representation as the pipeline default, especially for the automatic experiments where every additional input dimension also imports additional pose-estimation noise. This connects back to the representation discussion (Section 6.3): both results favour parsimony. Reducing the input to a small, well-chosen set of joints limits the model's exposure to noise and to spurious appearance variation, which is advantageous precisely because the dataset is small and the upstream estimates are imperfect. The choice of features should therefore be coupled to the data regime rather than maximised in isolation.

6.7 Effect of Frame Resolution per Cycle

The frames-per-cycle ablation (Section 5.9) showed no benefit from increasing temporal resolution beyond the eight-phase representation, and there was no monotonic trend across frame counts. The eight-phase sampling remained the strongest and most compact setting. The interpretation is that eight phases already capture the salient kinematic structure of a running cycle for this binary task; finer sampling mostly adds redundant frames and, on limited data, additional opportunity to overfit rather than additional discriminative signal.

This finding has a useful methodological consequence: the eight-phase representation is justified not merely by convention but by an explicit cost-benefit comparison. It keeps sample dimensionality low, which is consistent with the parsimony favoured by the model and feature analyses, and it reduces the computational burden of the pipeline without a meaningful accuracy penalty.

6.8 Cycle-Level versus Video-Level Interpretation

Aggregating cycle predictions to the video level by majority vote consistently improved the automatic baseline (Section 5.13), and the same direction held for the other pipelines. This is the expected behaviour when per-cycle predictions contain independent noise: aggregating several cycles from the same video lets correct predictions outweigh sporadic errors, reducing variance. Because the underlying data unit is the video, not the cycle, video-level reporting is also the more faithful measure of practical performance and is therefore the appropriate headline metric for any future deployment-oriented evaluation.

The interpretation must nonetheless be tempered by sample size. The video-level evaluation rests on only a handful of test videos, several of which contribute a single cycle. The improvement from aggregation is consistent and theoretically motivated, but its precise magnitude is uncertain at this scale, and the result should be read as directional evidence rather than as a calibrated estimate.

6.9 Effect of Threshold Selection

The threshold analysis (Section 5.14) revealed an asymmetry between the clean and the automatic pipelines. On the handmade data, validation-selected thresholds were mostly neutral or beneficial. On the automatic data, the same procedure left the robust MLP unchanged but sharply degraded the sequential models, with the CNN-BiLSTM losing substantial macro-F1 and being flagged as threshold-overfit.

This is the resolution of the apparent contradiction with the model ablation: the CNN-BiLSTM looks best at the default threshold but is unstable once a validation-chosen threshold is applied to test, whereas the MLP is stable, which is why it is the reported baseline. The deeper interpretation is that decision-threshold tuning is only as reliable as the validation set on which it is performed. With a small validation split and noisy automatic cycles, the validation-optimal threshold need not transfer, and tuning can transfer noise rather than signal.

The methodological implication, reflected in the reporting throughout Chapter 5, is that any tuned result must state that the threshold was selected on validation and must be accompanied by the default-threshold result, so that threshold overfitting is visible rather than hidden.

6.10 Effect of Pose and Cycle Quality

The quality analysis (Section 5.15) provides the most direct evidence for the segmentation-bottleneck interpretation. Classification accuracy on the automatic pipeline rose with higher heel visibility, lower landmark missing-ratio, and stronger autocorrelation of the cycle signal. In other words, the errors are not distributed uniformly across the test set; they concentrate in cycles where pose estimation is unreliable or the periodic structure is weak.

Together with the failed-video breakdown, which attributes the majority of failures to the cycle-detection stage, this locates the dominant error source firmly upstream of the classifier. This is the unifying thread of the discussion. The cycle-source gap (Section 6.2), the fragility of expressive models under automatic cycles (Section 6.4), the preference for the boundary-precise `contact_diff` signal (Section 6.5), and the instability of the image pipeline (Section 6.3) are all consequences of the same underlying limitation: imperfect markerless pose estimation and the difficulty of detecting clean running cycles in uncontrolled footage. The quality analysis makes that limitation measurable.

6.11 Error Analysis

Synthesising the error evidence, three failure modes can be distinguished. First, *detection failures*: videos in which no usable cycle is found, so the system cannot make a prediction at all; these dominate the failed-video counts and arise mainly at the cycle-detection stage (Section 5.15) . Second, *segmentation errors*: cycles that are detected but poorly bounded, producing misaligned eight-phase samples that degrade classification even when pose visibility is adequate; the validation-to-test collapse of the sequential models is the clearest symptom (Section 5.3) . Third, *classifier errors* on genuinely clean cycles, which the handmade results show to be comparatively rare.

The relative weight of these modes is itself a result: the first two, both upstream, account for the bulk of the gap between the upper bound and the automatic baseline, while the third is small. This ordering should guide remediation. Reducing detection and segmentation errors — through more robust pose estimation, better signal conditioning, or a learned boundary detector trained on more data — is likely to yield far larger gains than further work on the classifier. The exploratory boundary-detector and raw-sequence pipelines (Section 5.16) are first steps in that direction, but on the present data they did not yet improve over the rule-based baseline and remain inconclusive.

6.12 Limitations

Several limitations bound the conclusions of this work. The dataset is small, and the validation and test splits in particular are very small (Section 5.1) ; consequently, single-point differences, figures after threshold tuning, and the highest-scoring feature configuration should be treated as indicative rather than precise. The 5-fold fair-gap check partly mitigates this concern by repeating the main handmade–autocorr contrast across folds, but it does not remove the need for a larger external dataset. The video-level results also rest on few videos. The class distribution is moderately imbalanced at the video level, which is mitigated but not eliminated by the use of macro-F1 [22].

The pipeline also inherits the limitations of its components. Markerless pose estimation is itself noisy on in-the-wild footage [7]. Recent markerless-running analyses also note that such noise can affect downstream biomechanical interpretation [3]. In this pipeline, that noise propagates into signal extraction, cycle detection, and — for the image pipeline — crop stability.

Cycle detection relies on a periodicity assumption that does not hold for very short clips or atypical gait, contributing to detection failures. The handmade pipeline, used as the upper bound, depends on manual segmentation and therefore reflects an idealised condition that an automatic system cannot fully attain. Finally, all results are obtained on a single corpus collected under one labelling convention; generalisation to other populations, camera viewpoints, and definitions of "correct" technique remains untested. The historical version-history results (Section 5.10) are excluded from all benchmark claims for the reasons given above and add a further caution against over-reading development-stage numbers. The permutation test is also limited by the same small autocorr sample size and should be interpreted as a diagnostic robustness check, not as a definitive statistical claim about all possible classifiers.

6.13 Practical Implications

The practical implications follow directly from the bottleneck analysis. For researchers and practitioners building on this work, effort is best invested in the upstream stages: more robust markerless pose estimation, signal conditioning that improves cycle-boundary precision, and data-driven boundary detection trained on enough examples to outperform rule-based autocorrelation. The evidence indicates that such improvements would raise the automatic pipeline toward the handmade upper bound more effectively than any change to the classifier or any increase in temporal or feature resolution. Where an automatic system is used, predictions should be reported and consumed at the video level, with explicit handling of videos for which no reliable cycle can be detected.

It must be stated plainly that these results do not establish deployment readiness. The automatic pipeline performs near chance on the controlled test set, the evaluation rests on a small single-source dataset, and the upper bound depends on manual segmentation that a deployed system would not have.

The appropriate framing of the contribution is diagnostic: the work demonstrates that markerless pose sequences can encode running-technique distinctions when cycles are clean, and it identifies, through controlled comparison and ablation, exactly where an automatic in-the-wild system currently breaks down. Turning that diagnosis into a usable tool is left to future work (Chapter 7) .

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This thesis addressed the problem of classifying running technique as correct or false from *in-the-wild* internet videos using markerless pose estimation and deep learning. The work developed a complete pipeline consisting of markerless pose estimation, landmark filtering and normalisation, motion-signal extraction, running-cycle detection, eight-phase sampling, landmark-based or image-based cycle representation, deep-learning classification, and evaluation under a fixed split by video identifier.

The experimental design separated two questions that are often mixed together: whether the pose representation is learnable when cycle boundaries are reliable, and how much performance is lost when cycle boundaries must be detected automatically. The handmade landmark pipeline, using controlled cycle segmentation, reached a test accuracy of 0.9107 and a macro-F1 of 0.8991. This result is interpreted as an upper bound, not as a deployment baseline. The autocorrelation-based landmark pipeline, using automatic cycles detected from the `contact_diff` signal, reached a test accuracy of 0.5455 and a macro-F1 of 0.5417 under the reported baseline setting. Because the two pipelines use the same general pose-representation family but differ in cycle source, the gap between them localises the main difficulty of the task in the automatic cycle-detection stage. This interpretation is supported by an additional fair-gap robustness check: under the same MLP classifier and 5-fold video-level cross-validation, handmade cycles achieved 0.860 ± 0.016 macro-F1, whereas autocorr cycles achieved 0.615 ± 0.081, giving a fair gap of 0.245 macro-F1. A 100-permutation test on the autocorr MLP setting did not reach statistical significance ($p = 0.2475$), suggesting that the autocorr-derived features are not yet robustly separable under that simple classifier.

The ablation studies and extended analyses refine this conclusion. The model ablation showed that architecture choice matters less than segmentation quality: the CNN-BiLSTM is strongest on clean handmade cycles, while the simpler MLP is more stable as the reported automatic baseline after validation-based threshold selection. The feature ablation showed that compact selected-landmark representations are sufficient and competitive with larger landmark sets. The frames-per-cycle ablation showed that eight-phase sampling is an efficient choice, since increasing the number of sampled frames did not improve macro-F1.

The signal ablation showed a trade-off between detection coverage and semantically meaningful foot-strike-to-foot-strike boundaries; `contact_diff` was therefore retained as the main baseline signal even though `hip_y` achieved the highest classification macro-F1 in the signal ablation. The controlled landmark-versus-image comparison showed that the RGB-crop MobileNetV2–LSTM pipeline did not reproduce the historical image-based development result and was not stable enough to replace the landmark baseline.

Taken together, the results support a diagnostic conclusion. Markerless pose sequences can encode correct-versus-false running-technique differences when cycles are clean, but the current automatic pipeline is limited by pose quality, cycle detection, and boundary alignment. The fixed-split result and the cross-validation robustness check point to the same conclusion, while the permutation test shows that the autocorr representation remains statistically fragile under a simple MLP. The thesis therefore does not claim real-world deployment readiness. Its contribution is to identify where the automatic in-the-wild pipeline succeeds, where it fails, and which components should be improved before any practical feedback system is attempted.

7.2 Answers to Research Questions

The seven research questions introduced in Section 1.3 are answered below using the controlled results from Chapter 5.

RQ1 — Can pose-based classifiers distinguish correct from false technique under reliable segmentation? Yes. Under reliable manually controlled cycle segmentation, the handmade landmark upper bound reached a test accuracy of 0.9107 and a macro-F1 of 0.8991. All handmade models exceeded 0.85 macro-F1, and the best CNN-BiLSTM model reached the strongest overall result. This shows that the markerless pose representation contains useful information for distinguishing correct from false running technique when the running cycles are cleanly segmented.

RQ2 — How much performance is lost when handmade cycles are replaced by automatic autocorrelation-based detection? The loss is substantial. On the controlled fixed split, the reported autocorr landmark baseline reached a test accuracy of 0.5455 and a macro-F1 of 0.5417, compared with 0.9107 accuracy and 0.8991 macro-F1 for the handmade upper bound. This corresponds to a drop of 0.3652 in accuracy and 0.3574 in macro-F1. The fair 5-fold video-level CV check confirms that this loss is not only a fixed-split artefact: under the same MLP classifier and threshold 0.5, handmade cycles reached 0.860 ± 0.016 macro-F1, while autocorr cycles reached 0.615 ± 0.081, leaving a fair gap of 0.245 macro-F1. A permutation test on the autocorr MLP setting did not reach statistical significance ($p = 0.2475$), suggesting that autocorr-derived features are not yet robustly separable under this simple classifier. Since the comparison keeps the general landmark-representation family fixed while changing the cycle source, the loss is attributed primarily to automatic cycle detection and boundary alignment.

RQ3 — Which motion signal is most reliable for automatic cycle detection? The answer depends on the criterion. In the signal ablation, `hip_y` achieved the highest downstream classification macro-F1 (0.7321) and accuracy (0.7333). However, `contact_diff` was retained as the main automatic baseline signal because it is more directly tied to foot-contact dynamics and supports foot-strike-to-foot-strike cycle interpretation. Thus, `hip_y` was the strongest signal by classification score in this ablation, while `contact_diff` was the most appropriate baseline signal for semantically aligned cycle detection.

RQ4 — Which architecture is most suitable for eight-phase pose sequences? The answer is pipeline-dependent. On the handmade upper-bound pipeline, the CNN-BiLSTM achieved the best macro-F1 (0.8991), indicating that local temporal patterns and sequential modelling are useful when cycle alignment is reliable. On the automatic autocorr pipeline, the CNN-BiLSTM was strongest at the default 0.5 threshold, but it became unstable under validation-tuned thresholding. The MLP was therefore used as the more robust reported automatic baseline. This suggests that higher model capacity is useful only when the cycle samples are reliable; it does not compensate for noisy automatic segmentation.

RQ5 — How do feature representation and frames-per-cycle affect performance? The selected 13-landmark representation and the selected-landmark-plus-velocity variant matched the best handmade macro-F1 (0.8991), while the full 33-landmark and lower-body-only variants remained close but did not improve the overall macro-F1. This supports using a compact landmark representation as the default. For temporal resolution, the eight-frame setting achieved the best macro-F1 (0.8991), while 16, 24, and 32 frames did not yield a monotonic improvement. The eight-phase representation is therefore both effective and computationally economical for this dataset.

RQ6 — How does the image-based representation compare with the landmark-based representation under the autocorr pipeline? Under the controlled matched-cycle comparison, the landmark representation was clearly more reliable. The autocorr landmark baseline reached a macro-F1 of 0.5417, while the image-based MobileNetV2–LSTM pipeline reached only 0.1000 macro-F1 and a ROC-AUC of 0.0592. The historical `autocorr_fixed` image-based milestone with validation accuracy 76.5% was not reproduced under the final fixed-split protocol. It is therefore treated as development evidence and motivation for the organised image-based rerun, not as a benchmark result.

RQ7 — How do cycle aggregation, threshold selection, and pose/cycle quality affect interpretation? Each factor materially affects interpretation. Aggregating cycle-level predictions to the video level by majority vote improved the automatic landmark baseline from 0.5139 to 0.5333 macro-F1, although the video-level test set was small. Validation-tuned thresholds improved or preserved some clean handmade results but overfit several automatic-pipeline models, especially the CNN-BiLSTM. The quality analysis showed that higher heel visibility, lower missing-ratio, and stronger autocorrelation were associated with better classification accuracy. These findings confirm that evaluation should report cycle-level and video-level behaviour, state whether thresholds are tuned on validation, and account for pose and cycle quality.

7.3 Future Work

The findings point to several future directions, ordered by their likely impact on the automatic pipeline.

- **Larger and more diverse dataset.** The validation and test sets are small, so future work should collect more in-the-wild videos across more runners, camera viewpoints, running speeds, clothing conditions, and recording environments.
- **Stronger foot-strike and cycle-boundary annotation.** Since the largest performance gap comes from replacing handmade cycles with automatic cycles, higher-quality foot-strike annotations should be collected to train and evaluate better boundary detectors.
- **Improved automatic cycle detection.** The current autocorrelation approach is interpretable but limited. Future work should compare it with learned boundary detectors, temporal segmentation models, and hybrid methods that combine periodicity, foot-contact cues, and confidence-based filtering.
- **Pose-quality-aware filtering and weighting.** The quality analysis indicates that errors concentrate in low-visibility or weakly periodic cycles. Future systems should explicitly use pose visibility, landmark missing-ratio, autocorrelation strength, and cycle-quality scores to reject, weight, or calibrate predictions.
- **Expert biomechanics validation.** The correct/false labels and recovered kinematic patterns should be checked against expert assessment so that the classification target is better grounded in running-form criteria.
- **More robust image-based modelling.** The controlled image rerun showed that RGB crops are unstable in the current implementation. Future work should improve runner tracking, bounding-box smoothing, crop stabilisation, and data augmentation before drawing conclusions about the value of image-based representations.

- **Pose–image fusion.** Rather than choosing between landmarks and crops, future models could fuse a compact landmark stream with a stabilised visual stream, combining the robustness of pose coordinates with visual cues such as silhouette and limb contour.
- **Video-level decision protocol.** Since aggregation can reduce cycle-level noise, future work should treat video-level prediction as a first-class evaluation target, compare majority vote with probability averaging and confidence voting, and report uncertainty when few cycles are available.
- **Deployment only after validation.** A real-time coaching or feedback tool is a possible long-term direction, but it should only be considered after the automatic pipeline is validated on larger, representative data and shown to be reliable across viewpoints and runner populations.

The final conclusion is therefore deliberately conservative. This thesis shows that markerless pose data from in-the-wild running videos can support running-technique classification when cycle segmentation is reliable. It also shows that the current automatic system is limited mainly by upstream pose and cycle quality. The next stage of work should therefore prioritise robust cycle detection, larger validation data, and expert-labelled evaluation before moving toward practical deployment.

Appendix A

Implementation Details

This appendix documents the implementation behind the experiments of Chapters 4–5: the notebook organisation, the output directory layout, the files each stage produces, and the configuration settings. It is provided for reproducibility; no additional result claims are introduced here. Configuration values are reported only when they are available from the project notebooks or exported reports; values that were not exported are explicitly marked as unavailable rather than inferred.

A.1 Notebook Mapping

The pipeline is implemented as a sequence of self-contained notebooks. Table A.1 maps each notebook to the pipeline stage it realises and the principal role it plays in the thesis. Notebooks numbered with a letter suffix (03b–03g) are extended or supplementary analyses; the others are core pipeline stages.

A.2 Directory Structure

All notebooks write to a fixed output directory under a configurable `DRIVE_BASE` root, so that downstream aggregation notebooks can locate artefacts regardless of where the project is mounted. The principal layout is shown below.

```
DRIVE_BASE/
  outputs/
    dataset/          # split lists, split_summary, metadata
    eda/              # class/duration/fps/frame distributions, split charts
    handmade/         # handmade results, predictions, reports, curves, *.
   pth
    autocorr/         # autocorr results, cycle stats, fair-gap CV/
   permutation files
    autocorr_image/   # image results, cycle stats, crops, reports, *.pth
```

Table A.1: Mapping of project notebooks to pipeline stages and thesis roles.

ID	Notebook	Purpose	Role
00	`00_dataset_split_and_eda`	Dataset construction, video-level splitting, and exploratory data analysis.	Core
01	`01_handmade_8frame_upper_bound`	Handmade eight-phase cycle pipeline used as the controlled upper-bound reference.	Core
02	`02_autocorr_cycle_detection_baseline`	Automatic autocorrelation-based landmark baseline using motion signals and detected cycles.	Core
03	`03_handmade_vs_autocorr_comparison`	Comparison between manually controlled cycle segmentation and automatic cycle detection.	Core
03b	`03b_autocorr_image_based_pipeline`	Image-based autocorr pipeline using runner crops and a MobileNetV2–LSTM classifier.	Supplementary
03c	`03c_autocorr_version_comparison`	Autocorr version history and method-evolution summary.	Supplementary
03d	`03d_representation_comparison_landmark_vs_image`	Landmark-only versus image-based representation comparison.	Supplementary
03e	`03e_cycle_level_vs_video_level_evaluation`	Cycle-level and video-level evaluation with aggregation.	Supplementary
03f	`03f_pose_cycle_quality_analysis`	Pose-quality and cycle-quality analysis for interpreting failures.	Supplementary
03g	`03g_threshold_05_vs_tuned_report`	Comparison between default threshold 0.5 and validation-tuned threshold.	Supplementary
03h	`03h_autocorr_cv_robustness_check`	Fair handmade–autocorr gap under 5-fold video-level CV and autocorr MLP permutation test.	Core support
04	`04_model_ablation`	Architecture ablation over MLP, 1D-CNN, BiLSTM, and CNN-BiLSTM.	Core
05	`05_feature_ablation`	Input feature-set comparison.	Core
06	`06_signal_ablation_cycle_detection`	Comparison of motion signals for cycle detection.	Core
07	`07_frames_per_cycle_ablation`	Comparison of 8, 16, 24, and 32 sampled frames per cycle.	Core
08	`08_boundary_detector_pipeline`	Learned boundary-detection pipeline.	Supplementary
09	`09_raw_sequence_bilstm_gru`	Raw-sequence BiLSTM/GRU pipeline without fixed eight-phase sampling.	Supplementary
10	`10_threshold_tuning_and_calibration`	Threshold sweep and calibration analysis.	Supplementary
11	`11_error_analysis`	Error analysis used to support the discussion of failure modes.	Core support
12	`12_final_report_tables_and_figures`	Core report tables and figures.	Core
12b	`12b_extended_final_report_tables`	Extended report tables and final summary figures.	Supplementary

```
ablation/          # model / feature / signal / frames ablation CSVs
reports/           # comparison CSVs/MD, extended summaries, figures
figures/           # copied thesis-ready figures
```

A.3 Output Files

Table A.2 lists the principal machine-readable outputs and the notebook that produces each. These files are the inputs to the controlled comparisons and ablations of Chapter 5.

Table A.2: Principal output files and producing notebooks.

File	Producing notebook	Contents
`dataset/split_summary.csv`	00	Per-split video and cycle counts.
`handmade/handmade_results.csv`	01	Handmade upper-bound metrics for the landmark models.
`handmade/handmade_predictions.csv`	01	Per-sample predictions for the handmade upper-bound pipeline.
`handmade/handmade_classification_report.txt`	01	Classification report for the handmade pipeline.
`autocorr/autocorr_results.csv`	02	Autocorr landmark baseline metrics.
`autocorr/autocorr_predictions.csv`	02	Per-sample predictions for the autocorr landmark pipeline.
`autocorr/autocorr_cycle_stats.csv`	02	Per-cycle detection statistics and quality information.
`autocorr/autocorr_failed_videos.csv`	02	Failed or rejected videos in the autocorr pipeline.
`autocorr/fair_gap_and_mlp_perm.csv`	03h	Fair-gap CV summary and autocorr MLP permutation-test values.
`autocorr/fair_gap_and_mlp_perm.json`	03h	Machine-readable fair-gap and permutation-test record.
`reports/pipeline_comparison.csv`	03	Handmade versus autocorr comparison.
`autocorr_image/image_results.csv`	03b	Image-based autocorr pipeline metrics.
`autocorr_image/image_predictions.csv`	03b	Per-sample predictions for the image-based pipeline.
`autocorr_image/image_cycle_stats.csv`	03b	Cycle and crop statistics for the image pipeline.
`reports/autocorr_version_comparison.csv`	03c	Autocorr version history and method-evolution records.
`reports/representation_comparison.csv`	03d	Landmark versus image representation comparison.
`reports/cycle_level_metrics.csv`	03e	Cycle-level evaluation metrics.
`reports/video_level_metrics.csv`	03e	Video-level aggregated evaluation metrics.
`reports/pose_cycle_quality_metrics.csv`	03f	Quality-group metrics for pose and cycle quality analysis.
`reports/threshold_05_vs_tuned.csv`	03g	Default-threshold versus tuned-threshold comparison.
`ablation/model_ablation.csv`	04	Architecture ablation results.
`ablation/feature_ablation.csv`	05	Feature-set ablation results.
`ablation/signal_ablation.csv`	06	Motion-signal ablation results.
`ablation/frames_ablation.csv`	07	Frames-per-cycle ablation results.
`bd_results.csv`	08	Boundary-detector pipeline results, when exported.
`raw_sequence_results.csv`	09	Raw-sequence BiLSTM/GRU results, when exported.
`reports/main_pipeline_comparison.csv`	12b	Consolidated main pipeline summary.
`reports/extended_final_results_summary.csv`	12b	Extended final summary including supplementary pipelines.

A.4 Hyperparameter Settings

Table A.3 records implementation settings that could be verified from the project notebooks, exported reports, and the hyper-parameter evidence audit. The table intentionally separates settings that were verified from settings that were not exported. This prevents the thesis from inventing training details while still documenting the values that are available in the experiment evidence.

A.5 Reproducibility Notes

The experiments use video-level splitting to prevent cycles extracted from the same source video from appearing in both training and evaluation partitions. The handmade pipeline should be interpreted as an upper-bound reference because the cycle boundaries are manually controlled. The autocorr landmark pipeline is the main automatic baseline because it uses automatic cycle detection and the fixed split used for the final thesis results. The image-based pipeline is an extended representation comparison and should not be conflated with historical image-based development notebooks unless rerun under the same fixed-split protocol.

Table A.3: Verified implementation settings and model-configuration evidence.

Setting	Verified value / status
Pose estimator	MediaPipe BlazePose, producing 33 pose landmarks with landmark visibility.
Selected landmarks	13 landmarks: 0, 11, 12, 23, 24, 25, 26, 27, 28, 29, 30, 31, and 32.
Cycle representation	Eight sampled phases per detected running cycle.
Velocity augmentation	Enabled for the landmark representation where selected landmark coordinates are augmented with temporal velocity features.
Main automatic cycle signal	`contact_diff`.
Autocorr cycle logic	Foot-strike-to-foot-strike cycle detection with two-stride correction.
Autocorr visibility threshold	0.45.
Autocorr cycle duration bounds	$[0.50, 1.15]$ seconds for the controlled landmark baseline; the historical image-development notebook used a wider automatic-detection range before stricter cycle filtering.
Autocorr fixed margin	Disabled; `USE_MARGIN = False`.
Image crop size	224×224 RGB runner-centred crop.
Image backbone	MobileNetV2 visual feature extractor with an LSTM temporal head.
Landmark model training schedule	Verified from the audited landmark notebooks: learning rate 10^{-3}, batch size 16, 80 epochs, weight decay 10^{-4}, random seed 42. Optimiser and loss function were not consistently exported in the final evidence.
MLP parameter count	193,025 parameters.
1D-CNN parameter count	40,257 parameters.
BiLSTM parameter count	173,441 parameters.
CNN-BiLSTM parameter count	82,113 parameters.
Autocorr image representation schedule	Verified from the audited `03b_autocorr_image_based_pipeline.ipynb`: learning rate 10^{-3}, batch size 8, 40 epochs, weight decay 10^{-4}, dropout 0.4, two recurrent layers, random seed 42.
Historical `autocorr_fixed` image-development model	Verified from the historical 76.5% notebook: 8 RGB frames per cycle, batch size 8, MobileNetV2 ImageNet backbone, LSTM hidden size 128, two LSTM layers, dropout 0.4, LayerNorm + two-layer classification head.
Historical `autocorr_fixed` training schedule	AdamW with OneCycleLR in two phases: 15 frozen-backbone epochs with lr 3×10^{-4} and weight decay 10^{-3}, followed by 10 partial-fine-tuning epochs with lr 5×10^{-5} and weight decay 10^{-4}. The last five MobileNetV2 feature blocks were unfrozen in the fine-tuning phase.
Historical `autocorr_fixed` loss and regularisation	`BCEWithLogitsLoss` with class `pos_weight`, gradient clipping at 1.0, validation accuracy used for checkpoint selection in the historical development notebook.
MobileNetV2–LSTM parameter count	*Not exported in provided reports.*
Boundary-detector evidence	Dropout values 0.3/0.4 and two recurrent layers were found in the audit, but the full training schedule was not consistently exported; the boundary detector is therefore treated as supplementary.
GRU / raw-sequence parameter count	*Not exported in provided reports.*
Raw-sequence training schedule	*Not exported in provided reports.*
Early stopping	*Not exported in provided reports.*

Note. Parameter counts for MLP, 1D-CNN, BiLSTM, and CNN-BiLSTM are taken from the exported model-ablation report. Landmark and image training schedules are reported only when they were explicitly recovered from the audited notebooks or the hyper-parameter evidence audit. The historical `autocorr_fixed` settings describe the development image pipeline that produced the 76.5% validation-accuracy milestone; they are documented as method evidence and should not be confused with a controlled benchmark result unless that pipeline is rerun under the shared train/validation/test protocol.

Appendix B

Additional Results

This appendix collects supplementary material referenced from Chapters 5–6: additional confusion matrices, cycle-detection signal examples, training curves, failed-case summaries, and the autocorrelation version history. No new controlled benchmark results are introduced here; the controlled results remain those reported in Chapter 5.

B.1 Additional Confusion Matrices

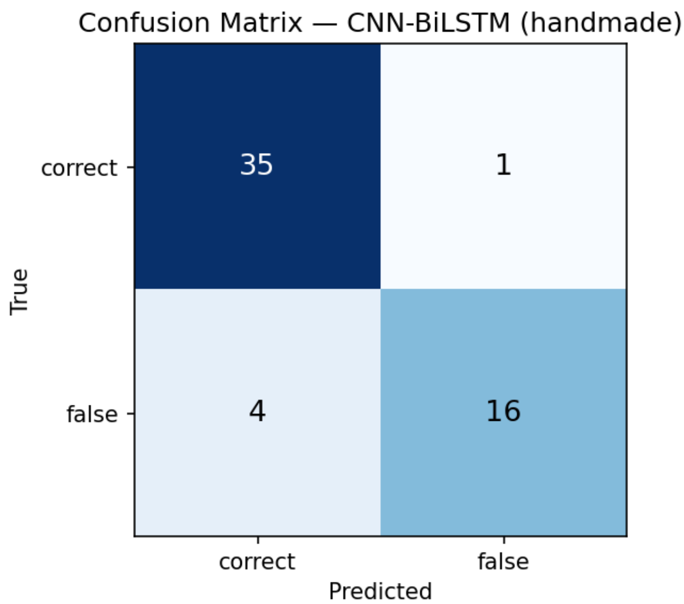


Figure B.1: Confusion matrix of the best handmade upper-bound model on the test set.

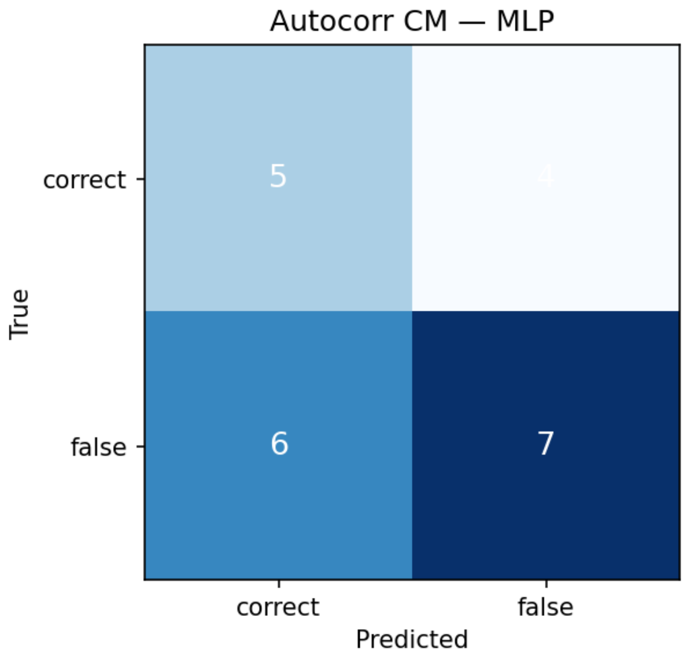


Figure B.2: Confusion matrix of the autocorr landmark baseline on the test set.

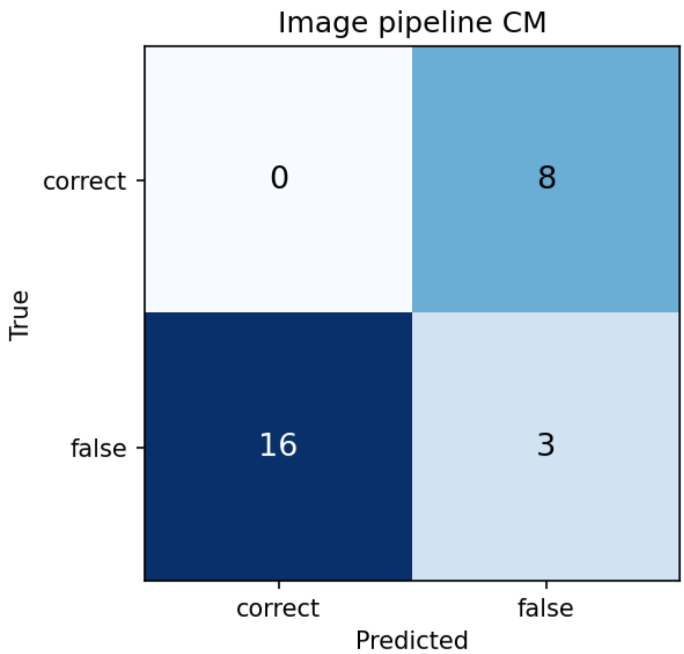


Figure B.3: Confusion matrix of the autocorr image-based pipeline on the test set.

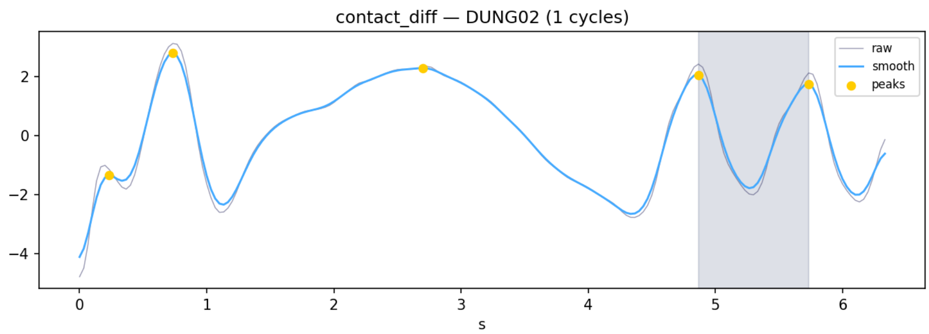


Figure B.4: Example `contact_diff` motion signal used for autocorrelation-based cycle detection.

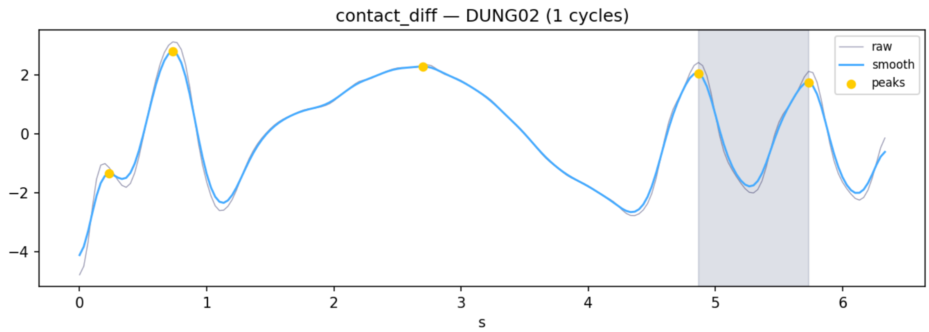


Figure B.5: Example of an automatically detected foot-strike-to-foot-strike running cycle.

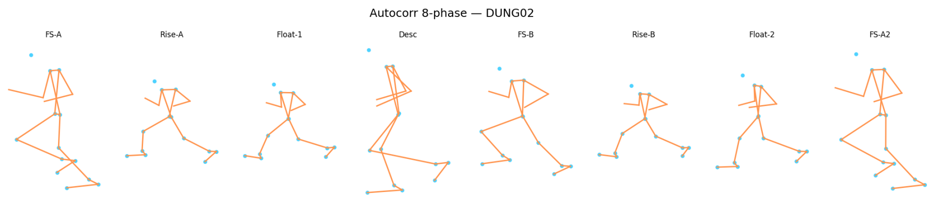


Figure B.6: Example eight-phase sampling of an automatically detected cycle.

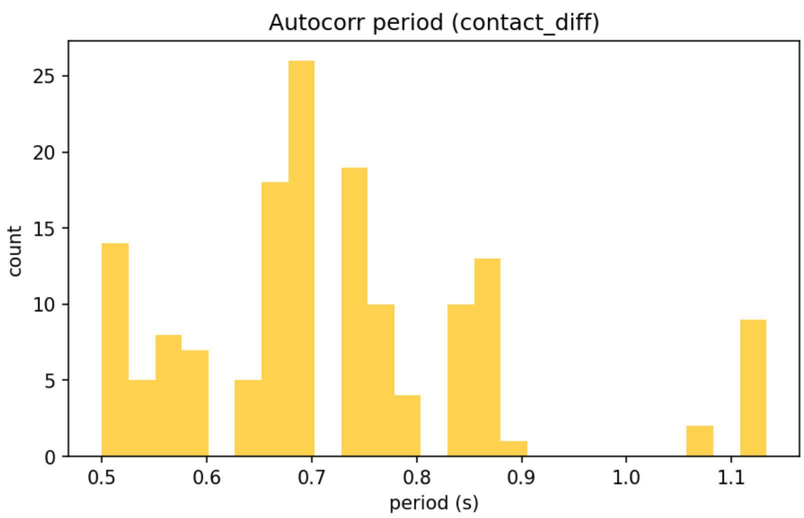


Figure B.7: Distribution of estimated cycle periods across detected videos.

B.2 Additional Signal Examples

B.3 Training Curves

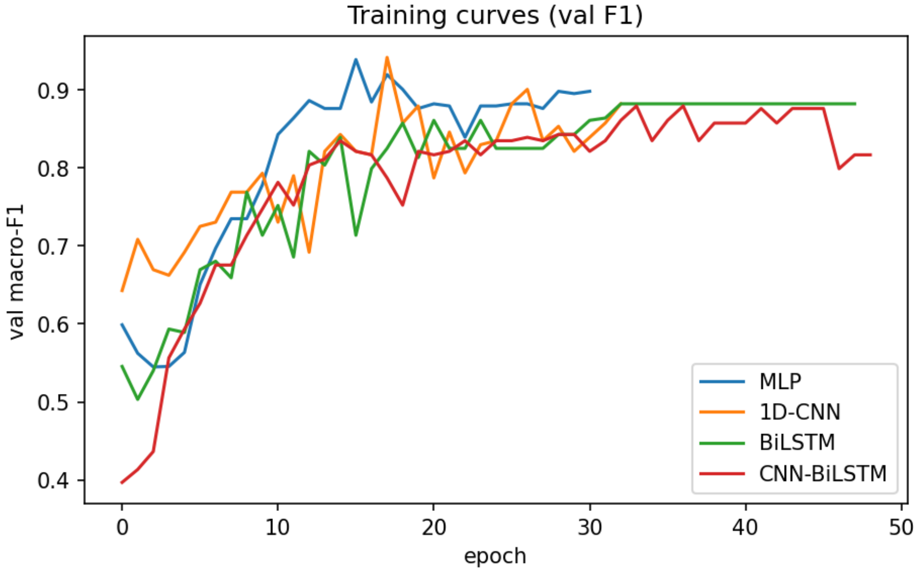


Figure B.8: Training and validation curves for the handmade upper-bound models.

Supplementary figure not available in the LaTeX tree: `figures/image_training_curve.png`.

B.4 Failed Case Examples

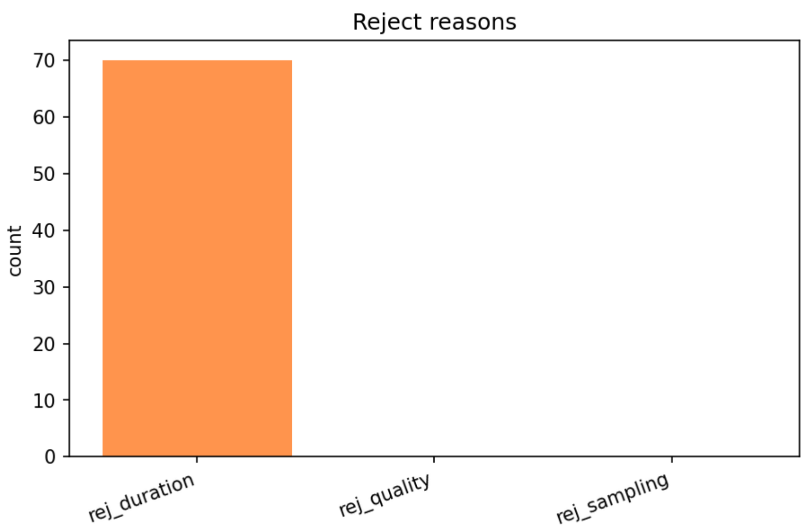


Figure B.9: Breakdown of cycle-level rejection reasons in the autocorr pipeline.

The per-video failure records are available in `outputs/autocorr/autocorr_` failed_videos.csv and the image-pipeline counterpart outputs/autocorr_image/failed_videos.csv, when those files are exported by the corresponding notebooks.

B.5 Fair Gap Robustness and Permutation Test

Table B.1 records the supplementary robustness check used in Section 5.5. It repeats the handmade–autocorr comparison with the same MLP classifier, a fixed threshold of 0.5, and 5-fold video-level cross-validation. The values support the fixed-split conclusion that automatic cycle detection is the main bottleneck.

Table B.1: Fair-gap robustness check and autocorr MLP permutation test. Source: outputs/autocorr/fair_gap_and_mlp_perm.json.

Quantity	Value
Handmade cycles / videos	370 / 369
Autocorr cycles / videos	151 / 55
Handmade 5-fold CV macro-F1	0.8597 ± 0.0160
Handmade OOF macro-F1, 95% CI	0.8599 [0.8219, 0.8969]
Autocorr 5-fold CV macro-F1	0.6146 ± 0.0809
Autocorr OOF macro-F1, 95% CI	0.6553 [0.5797, 0.7281]
Fair gap (handmade − autocorr)	0.2451
Observed autocorr MLP CV macro-F1 in permutation test	0.5206
Permutation null mean ± std.	0.4680 ± 0.0740
Number of permutations	100
Permutation p-value	0.2475

The permutation result is interpreted conservatively. It does not prove that the autocorr cycles contain no signal; it indicates that the signal is not robustly separable under the tested simple MLP configuration and current sample size.

B.6 Autocorr Version History

This section reproduces the autocorrelation version history for completeness. **These are historical development results, not controlled benchmarks.** The reported validation accuracies were recorded in development notebooks under varying protocols and are *not* directly comparable to the controlled test metrics of Chapter 5. In particular, autocorr_fixed(val_acc 76.5) is a historical image-based milestone that motivated the organised image-based rerun. It must not be used to claim that image-based modelling outperforms the landmark baseline unless rerun under the same fixed-split protocol.

Note. The historical validation values in Table B.2 are development-stage records and are not directly comparable to the fixed-split test metrics reported in Chapter 5.

Table B.2: Autocorr version history. These rows describe historical development stages, not controlled benchmark results. Validation accuracies are reported only when they were recorded in development evidence.

Version	Signal	Input	Model	Val. acc.
auto-corr_fixed_val_acc_69_6	head-Y oscillation	landmarks	CNN+LSTM / historical	69.6%
auto-corr_fixed_val_acc_72_9	`contact_diff`	landmarks	CNN+LSTM / historical	72.9%
autocorr_fixed_biLSTM	`contact_diff`	landmarks + velocity	BiLSTM	Not recorded
auto-corr_fixed_val_acc_76_5	`contact_diff` + autocorr period	RGB crop 224×224	MobileNetV2 + LSTM	76.5%

B.7 Historical Image-Based Autocorr Pipeline

Among the historical autocorrelation-based experiments, `autocorr_fixed(val_acc 76.5)` represents the most technically complex development stage. Earlier autocorr versions mainly relied on pose-landmark sequences and hand-crafted motion signals, such as head vertical oscillation or contact-related landmark differences. In contrast, this version extended the pipeline from landmark-based time-series classification to image-based sequence modelling.

The pipeline first used markerless pose estimation to localise the runner and extract foot-contact-related motion signals. The `contact_diff` signal, together with autocorrelation-based period estimation, was used to identify running cycles. Instead of using only normalised landmark coordinates as the input representation, each detected cycle was converted into an eight-phase RGB image sequence. For each sampled phase, a runner-centred crop was generated using a pose-derived bounding box, resized to a fixed image resolution, and processed by a MobileNetV2-based visual feature extractor. A temporal LSTM head then aggregated the eight-frame image sequence to classify the running technique as correct or false.

This experiment is important because it tested whether visual appearance cues that are absent from sparse landmark representations could improve classification. RGB crops may preserve information about body silhouette, limb contours, clothing-relative motion, and visual context that is lost when the input is reduced to pose coordinates. Therefore, this notebook should be described as a methodological extension from pose-only sequence classification toward image-based running-cycle classification.

However, `autocorr_fixed(val_acc 76.5)` should not be used as the main benchmark result. The reported 76.5% value is a historical validation accuracy from a development notebook and was not confirmed under the final fixed-split evaluation and unified reporting protocol. For this reason, it is interpreted as development evidence rather than final comparative evidence. The organised image-based rerun is reported separately in Chapter 5 as an extended representation comparison.

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